

One-to-Many Heterogeneous Identity Authentication for Privacy-Preserving Vehicular Service Computing

Zirui Qiao, Jing Wang, Guangjin Zhang, Yanwei Zhou, Yuxuan Liu, Baodong Qin, Dong Zheng

Abstract—The rapid development of vehicular ad hoc networks (VANETs) has increased the privacy risks associated with the transmission, processing, and storage of vehicle-generated data containing sensitive owner information. Conventional encryption mechanisms provide basic security guarantees. These guarantees alone are insufficient to support fine-grained access control in dynamic, large-scale VANET-based service environments. The exposure of plaintext after decryption further limits secure and precise service-side data processing and analysis. To address these challenges, this paper proposes an inner product functional encryption (IPFE)-based heterogeneous identity authentication scheme for VANETs. The scheme integrates identity-based cryptography with public key infrastructure through a heterogeneous identity authentication protocol, enabling authentication between mobile vehicles and certificate-based application servers. By leveraging IPFE, the scheme supports authorized computation over encrypted vehicular data without disclosing sensitive owner information, preserving data confidentiality and service-side processing capability. A one-to-many authentication mechanism further improves authentication efficiency, reduces communication overhead, and enhances real-time performance in large-scale vehicular service scenarios. The security of the proposed scheme is formally proven under standard assumptions. Comprehensive performance evaluations demonstrate favorable computational, storage, and communication efficiency. In addition, a simulation in a realistic vehicular environment (Veins/OMNeT++/SUMO) confirms that the one-to-many authentication keeps the end-to-end authentication latency bounded and reduces the server-side computation per authenticated vehicle as the fleet grows, while ciphertext aggregation bounds the decryption cost. These results indicate that the proposed scheme provides effective support for secure, scalable, and privacy-preserving intelligent transportation services.

Index terms— Inner-product function encryption, Heterogeneous communication, Aggregate encryption, Broadcast authentication.

I. INTRODUCTION

THE rapid proliferation of intelligent transportation systems and autonomous driving technologies has transformed vehicular ad hoc networks (VANETs) into service-oriented vehicular computing infrastructures, in which vehicles, roadside

units (RSUs), edge nodes, and cloud-based application servers (AS) continuously exchange data to support safety-critical, traffic-management, and intelligent mobility services [1]. This service-centric evolution increases data velocity and broadens the exposure surface of sensitive vehicular information. Conventional public-key encryption alone cannot simultaneously support fine-grained access control and privacy-preserving data utilization during service execution. Inner product functional encryption (IPFE) provides an effective cryptographic primitive by enabling the computation of inner products $\langle x, y \rangle$ over encrypted vectors x using keys associated with y . This capability enables ASs to perform secure data analysis without accessing plaintext, supporting privacy-preserving traffic optimization and service management.

The heterogeneous nature of VANET-based service computing also necessitates the integration of diverse cryptographic mechanisms. Public key infrastructure (PKI) provides robust authentication for relatively stable, resource-sufficient AS. Identity-based cryptography (IBC) is well suited to mobile vehicles because its certificate-free property reduces key-management overhead. Existing one-to-one authentication schemes [2], [3] exhibit limited scalability, resulting in increased latency and energy consumption in large-scale vehicular service environments.

To address these limitations, this paper proposes an efficient heterogeneous identity authentication protocol that integrates one-to-many authentication with aggregated encryption by combining PKI and IBC. The proposed framework enables secure computation at the AS, preserves vehicle anonymity, and improves communication efficiency in dynamic VANET environments. In the proposed construction, the vehicle side adopts IBC to eliminate certificate management, making it suitable for resource-constrained vehicular nodes. The AS side employs PKI, which relies on digital certificates and is appropriate for fixed, resource-sufficient service providers. The proposed scheme achieves fine-grained access control and efficient communication, reducing computational and storage overhead and improving network resource utilization in large-scale VANET-based service computing.

A. Related Works

This section reviews identity authentication, functional encryption, and heterogeneous cryptographic systems, with emphasis on their relevance to privacy-preserving vehicular service computing.

(1) **Identity Authentication.** Recent research on identity authentication seeks to balance strong security guarantees and

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practical efficiency. Jiang *et al.* [4] proposed a strong federated authentication scheme using password-based credentials to resist identity-server corruption. Tian *et al.* [5] achieved unlinkability in mobile edge computing through one-time pseudonyms. Wang *et al.* [6] enabled broadcast authentication in VANETs and reduced authentication overhead. He *et al.* [7] and Chen *et al.* [8] integrated blockchain with self-sovereign identity to support cross-domain traceability in the Internet of Things. Zhou *et al.* [9] employed physical unclonable functions to achieve lightweight and tamper-resistant authentication. Zeng *et al.* [10] leveraged an authenticated sponge construction to provide efficient security mechanisms for telehealthcare systems. Existing authentication schemes exhibit two major limitations. 1) **Limited support for heterogeneous cryptographic systems.** Many identity authentication methods are designed for a single cryptographic infrastructure, such as PKI, IBC, or certificateless cryptography. Interoperability between a PKI-based device and an IBC-based server requires reconciling distinct key formats, trust models, and identity-verification mechanisms. 2) **Limited performance and ciphertext utility.** Existing schemes mainly focus on message authentication and do not incorporate ciphertext-based computation into service-side processing. Service providers must therefore decrypt data before processing it, increasing the risk of sensitive information exposure. Most of the reviewed protocols, except [6], follow a one-to-one authentication model. Supporting group or broadcast authentication through repeated individual exchanges incurs communication and computational overhead that grows linearly with the number of participants.

Table I summarizes the comparative characteristics and limitations of existing authentication schemes. Let T_1 denote support for identity authentication across heterogeneous cryptographic systems, T_2 denote privacy-preserving computation over ciphertexts, T_3 denote one-to-many identity authentication, and T_4 denote the number of communication rounds required for identity authentication with n users. The symbol “✓” indicates that a scheme possesses the corresponding attribute, and “✗” denotes its absence. The symbol n represents the number of users participating in identity authentication.

TABLE I: Limitations of existing authentication schemes

Schemes	T_1	T_2	T_3	T_4	Limitations
Jiang <i>et al.</i> 's [4]	✗	✗	✗	n	Multi-device requires re-registration
Tian <i>et al.</i> 's [5]	✗	✗	✗	n	Online dependency and single-point-of-failure risk
Wang <i>et al.</i> 's [6]	✗	✗	✓	1	Single cryptographic system
He <i>et al.</i> 's [7]	✗	✗	✗	n	High computation overhead
Chen <i>et al.</i> 's [8]	✗	✗	✗	n	Requires pre-distributed keys
Zhou <i>et al.</i> 's [9]	✗	✗	✗	n	State synchronization
Zeng <i>et al.</i> 's [10]	✗	✗	✗	n	Insufficient resistance to insider attacks

(2) **Functional Encryption.** Functional encryption (FE), introduced by Boneh *et al.* [11], enables function-specific computation over encrypted data without revealing the underlying plaintext. A trusted authority generates the system parameters and issues function-specific secret keys. IPFE supports fine-grained access control by allowing a user holding a secret key sk_y to compute only $\langle x, y \rangle$ from an encryption of x ,

without disclosing x itself. This paradigm limits information disclosure and strengthens data privacy in practical service deployments. IPFE constructions with appropriate security guarantees can also resist active attacks and preserve confidentiality under adversarial interference. These properties make IPFE a suitable primitive for privacy-preserving service analytics. Dowerah *et al.* [12] designed an attribute-based FE scheme with hidden attributes and functions, together with scalable encryption. Kaminsky *et al.* [13] introduced FE-based cross-domain solutions for multi-level secure exchanges. Yang *et al.* [14] integrated FE with blockchain to optimize VANET access control and reliable transmission. Panzade *et al.* [15] leveraged function-hiding FE to secure neural-network activation computations and improve the efficiency of inner-product operations.

Current research on FE primarily focuses on cryptographic constructions for privacy-preserving computation, with particular emphasis on theoretical methods and security guarantees for computation over encrypted data. Comparatively limited attention has been devoted to integrating FE with practical application-layer protocols, leaving a gap between theoretical secure computation and its deployment in real-world service scenarios.

(3) **Heterogeneous Systems.** Heterogeneous cryptographic systems facilitate secure communication among entities operating under distinct cryptographic infrastructures. This approach allows each domain to select a cryptographic mechanism according to its service requirements, resource constraints, and trust assumptions.

In practical network deployments, heterogeneous cryptographic systems allow domains to retain their own key-management policies and establish secure cross-platform interactions. Users can select key-generation mechanisms consistent with their trust assumptions and computational capabilities. Appropriate interoperability mechanisms can reduce communication overhead, improve scalability in large-scale decentralized service environments, and preserve compatibility with existing infrastructures. Ali *et al.* [16] proposed a pairing-based hybrid signcryption scheme. Devi *et al.* [17] combined attribute-based encryption with blockchain-based aggregation. Ali *et al.* [18] developed a certificateless heterogeneous signcryption scheme. These studies primarily address fundamental cryptographic primitives and give limited attention to application-level protocol design and service-oriented requirements.

For heterogeneous identity authentication services, Cai *et al.* [19] proposed a cooperative authentication scheme for heterogeneous networks to enable privacy-preserving, cross-network device access. Iqbal *et al.* [20] presented a secure authentication and key-management framework for heterogeneous wireless sensor networks to achieve energy-efficient, mutually authenticated data transmission among nodes with heterogeneous capabilities. Zhou *et al.* [21] designed a lightweight cross-domain authentication and key-agreement protocol that enables secure mutual authentication across heterogeneous IoT domains. The heterogeneity considered in these studies encompasses differences in network standards and authentication protocols [19], sensor-node capabilities [20], and systems

across management domains [21]. These settings are distinct from the underlying cryptographic-system heterogeneity considered in this paper. Among the studies reviewed here, Xu *et al.* explicitly addressed differences between underlying cryptographic systems in identity authentication and proposed a scheme connecting traditional public key infrastructure with certificateless public-key cryptography [22]. However, all the authentication messages in the above scheme [22] should be fully decrypted before conducting the identity verification.

Overall, existing heterogeneous identity authentication schemes primarily address heterogeneity in terminal capabilities and network architectures, while the heterogeneity of underlying cryptographic systems has received comparatively limited attention. Existing studies also provide limited support for privacy-preserving computation at the underlying cryptographic level, leaving the integration of heterogeneous authentication and ciphertext-based data processing insufficiently explored.

B. Our Motivations

The increasing deployment of VANETs raises critical security and privacy concerns because of frequent exchanges of sensitive data among vehicles and service entities. Efficient authentication and privacy-preserving mechanisms are required to address the following challenges.

(1) **Efficient Authentication.** In large-scale VANET environments, vehicles continuously exchange safety-critical messages with RSUs, ASs, and other vehicles under strict latency constraints. Traditional one-to-one authentication schemes require repeated individual verification, imposing substantial computational and communication costs in dense traffic scenarios. Limited onboard resources and real-time safety requirements further intensify this challenge. Scalable mechanisms, such as batch verification and one-to-many authentication, are needed to reduce redundant cryptographic operations and maintain strong security guarantees.

(2) **Heterogeneous Communication.** In practical deployments, VANETs often span multiple administrative domains in which different cryptographic infrastructures, such as PKI and IBC, coexist. These infrastructures rely on distinct key-management and trust models, making secure interoperability a non-trivial task. Cross-domain service communication requires efficient mechanisms to bridge these differences without introducing excessive overhead or weakening security. A unified heterogeneous identity authentication protocol is therefore essential for secure interaction between PKI-based and IBC-based entities in flexible, large-scale vehicular service networks.

(3) **Information Aggregation.** In current VANET communication models, vehicles often transmit data independently to RSUs or cloud servers, generating numerous redundant messages over shared wireless channels. This traffic increases network congestion, degrades communication efficiency, and expands the exposure surface of sensitive information during transmission. The absence of effective aggregation mechanisms further limits scalability and increases privacy leakage risks. Secure aggregation of authentication credentials

or messages can reduce communication overhead and limit unnecessary data exposure.

C. Our Contributions

This paper proposes a heterogeneous identity authentication protocol that supports efficient and secure one-to-many authentication across heterogeneous cryptographic systems. The main contributions are summarized as follows.

(1) **Enabling statistical computation over encrypted data.** To avoid plaintext exposure during service-side analysis, the proposed protocol integrates IPFE into the authentication framework. This design enables the AS to evaluate general linear operations, including weighted averages, directly over encrypted vehicular data without fully decrypting the underlying vectors. It preserves the confidentiality of the underlying data and supports meaningful service-side computation, thereby protecting sensitive information throughout the analytical process.

(2) **Improving authentication scalability.** The proposed protocol addresses the authentication-efficiency bottleneck in broadcast-oriented service scenarios. It enables the AS to verify signatures from a large number of vehicles concurrently in VANET environments. A one-to-many identity authentication method reduces the number of communication rounds from $O(n)$ to $O(1)$, where n denotes the number of users involved in the identity authentication process. This reduction decreases repeated communication and improves authentication efficiency in large-scale vehicular service environments.

II. PRELIMINARIES

A. Notations

We have summarized the symbols used in our privacy protection protocol in Table II. Let $\mathbb{G}_1$, $\mathbb{G}_2$ and $\mathbb{G}_T$ be three multiplicative cyclic groups of prime order q , and let g be a generator of $\mathbb{G}_1$. Let $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ be an admissible bilinear map. For any vectors of length n , $\mathbf{x} = (x_1, x_2, \dots, x_n) \in (\mathbb{Z}_q^*)^n$ and $\mathbf{y} = (y_1, y_2, \dots, y_n) \in (\mathbb{Z}_q^*)^n$, we define $e(g, g)^{\mathbf{x}}$ as a valid computation, such that $\left(e(g, g)^{\mathbf{x}}\right)^{\mathbf{y}} = e(g, g)^{\langle \mathbf{x}, \mathbf{y} \rangle}$.

TABLE II: Definitions of notations used in this paper

Notations	Definitions
κ	Security parameter
V_i	The identity of vehicle
RSU_j	The identity of RSU
ID_A	The identity of AS
n	Dimension of vectors
$\langle \mathbf{x}, \mathbf{y} \rangle$	The inner-product of two vectors $\mathbf{x}$ and $\mathbf{y}$
$Params$	System public parameter
$ask_{\mathbf{y}}$	The private key of identity ID_A
vs_{k_i}	The private key of identity V_i
rs_{k_j}	The private key of identity RSU_j
apk	The public key of identity ID_A
$ct_{\mathbf{x}_i}$	The ciphertext corresponding to the V_i
$negl(\kappa)$	The negligible values on the security parameter κ

B. Complexity Assumptions

We assume that $\mathbb{G}_1$ and $\mathbb{G}_2$ are multiplicative cyclic groups of order q . Let $a, b \in \mathbb{Z}_q^*$, $T \in \mathbb{G}_1$ be randomly selected. Let g and h are the generators of $\mathbb{G}_1$ and $\mathbb{G}_2$, respectively. There are two distributions defined as $\mathcal{D}_1 = (g, g^a, g^b, h, g^{ab})$ and $\mathcal{D}_2 = (g, g^a, g^b, h, T)$.

The advantage of adversary $\mathcal{A}$ in solving External Diffie-Hellman (XDH) problem is defined as: $\text{Adv}_{\mathcal{A}}^{\text{XDH}}(\kappa) = |\Pr[\mathcal{A}(\mathcal{D}_1) = 1] - \Pr[\mathcal{A}(\mathcal{D}_2) = 1]|$.

Definition 1 (External Diffie-Hellman (XDH) Assumption): For any probability polynomial time (PPT) adversary $\mathcal{A}$, the XDH assumption holds if $\text{Adv}_{\mathcal{A}}^{\text{XDH}}(\kappa)$ is negligible.

The DL problem is defined as follows: Given a public tuple $(g, g^b) \in \mathbb{G}_1^2$, the task is to compute b , where $b \in \mathbb{Z}_q^*$, and the advantage of an adversary $\mathcal{A}$ in solving the DL problem is defined as: $\text{Adv}_{\mathcal{A}}^{\text{DL}}(\kappa) = \Pr[\mathcal{A}(g, g^b) = b]$.

Definition 2 (Discrete Logarithm (DL) Assumption): For any PPT adversary $\mathcal{A}$, the DL assumption holds if $\text{Adv}_{\mathcal{A}}^{\text{DL}}(\kappa)$ is negligible.

The CDH problem is defined as follows: Given a public tuple $(g, g^a, g^b) \in \mathbb{G}_1^3$, the task is to compute g^{ab} , where $a, b \in \mathbb{Z}_q^*$, and the advantage of an adversary $\mathcal{A}$ in solving the CDH problem is defined as: $\text{Adv}_{\mathcal{A}}^{\text{CDH}}(\kappa) = \Pr[\mathcal{A}(g, g^a, g^b) = g^{ab}]$.

Definition 3 (Computational Diffie-Hellman (CDH) Assumption): For any PPT adversary $\mathcal{A}$, the CDH assumption holds if $\text{Adv}_{\mathcal{A}}^{\text{CDH}}(\kappa)$ is negligible.

C. Lagrange Interpolation Polynomial

Given a function $y = f(x)$ with $n + 1$ known points: $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, we can construct a polynomial $L(x)$ of degree at most n defined as: $L(x) = \sum_{i=0}^n y_i l_i(x)$, which satisfies the condition $L(x_i) = y_i, i = 0, 1, \dots, n$, where $l_i(x)$ is the Lagrange fundamental polynomial given by $l_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$, and is denoted by $\Delta_{i,n}(x)$.

In our construction, Lagrange interpolation is employed to support threshold-based generation of vehicle private keys from partial private keys issued by multiple authorized RSUs. During system initialization, the CA randomly selects a polynomial $f_1(x) = s + a_1x + a_2x^2 + \dots + a_{\eta-1}x^{\eta-1}$ of degree $\eta - 1$ over $\mathbb{Z}_q$, where $f_1(0) = s$ is the CA master secret. Each authorized RSU RSU_j is associated with a distinct nonzero field element and obtains a private share $rsk_j = f_1(RSU_j)$. When vehicle V_i requests registration, RSU_j generates the corresponding partial private key $sk_{j,i} = H_1(V_i)^{c_i \cdot rsk_j}$, where $c_i = H(K_i, V_i)$. After obtaining valid partial private keys from any set $\mathcal{S}$ of at least η authorized RSUs, V_i computes the Lagrange coefficient $\Delta_{j,\mathcal{S}}(0) = \prod_{k \in \mathcal{S}, k \neq j} \frac{RSU_k}{RSU_k - RSU_j}$ for each participating RSU. By the Lagrange interpolation relation $\sum_{j \in \mathcal{S}} f_1(RSU_j) \Delta_{j,\mathcal{S}}(0) = f_1(0) = s$, the vehicle combines the received partial private keys as $vs_{ki} = H_1(V_i)^{k_i}$. $\prod_{j \in \mathcal{S}} sk_{j,i}^{\Delta_{j,\mathcal{S}}(0)} = H_1(V_i)^{k_i + c_i s}$. Therefore, Lagrange interpolation is performed implicitly in the exponent during vehicle private-key generation, rather than explicitly reconstructing the CA master secret s . This threshold mechanism ensures that no individual RSU can independently generate the complete

private key of a vehicle, while a sufficient set of authorized RSUs enables the vehicle to derive its valid private key.

III. SYSTEM MODEL AND SECURITY REQUIREMENTS

Resource-constrained vehicles employ IBC, in which private keys are generated by a private key generator (PKG). The resource-sufficient AS operates under PKI, with certificates issued by a certificate authority (CA). A single cryptographic framework is not well suited to the distinct key-management requirements of these two paradigms. Enforcing PKI on vehicles would impose substantial computational and storage overhead associated with certificate revocation list management. Enforcing IBC on the AS would introduce key-escrow risks by granting the PKG control over AS private keys. The proposed architecture therefore combines the two infrastructures: the PKG supports the IBC system for vehicles, and the CA manages the PKI system for the AS.

A. System Model

Fig. 1 illustrates the VANET system model. The model comprises five primary entities: the CA for PKI, the PKG for IBC, roadside units (RSUs), the AS, and vehicles. The functions of these entities are described as follows.

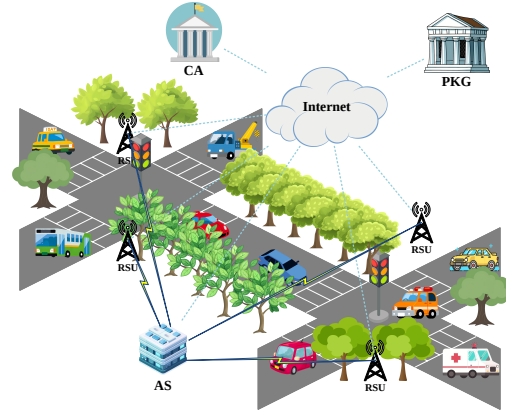


Fig. 1: System model of VANET

(1) **CA**. The CA establishes the PKI system. It generates the master private key and the corresponding public parameters and distributes shares of the master private key to the RSUs.

(2) **PKG**. The PKG initializes the IBC system by generating the public parameters and the master secret key. It also issues private keys to the AS within the IBC framework.

(3) **RSU**. Each RSU receives a private key share from the CA and generates partial private keys for vehicles within its coverage area. The system contains l RSUs in total, and more than t RSUs must collaborate to generate partial private keys for a vehicle. RSUs also collect communication data from vehicles within their respective domains.

(4) **Vehicle**. Each vehicle obtains partial private keys from RSUs and reconstructs its full private key once more than t RSUs have contributed. After completing mutual authentication with the AS, the vehicle transmits its relevant data to the corresponding RSU.

(5) **AS**. The AS submits the information to be evaluated to the PKG and obtains the corresponding private key. After completing mutual authentication with vehicles, it collects aggregated vehicular data from RSUs across different regions.

B. Heterogeneous Identity Authentication in VANETs

The proposed heterogeneous identity authentication protocol and message transmission scheme comprises the following stages.

(1) **System Initialization**. The PKG and the CA establish the IBC and PKI frameworks, respectively. They generate and publish the corresponding public parameters to support subsequent cryptographic operations.

(2) **Entity Registration**. Vehicles and the AS complete their respective registration procedures with the assistance of the PKG and the CA. This stage generates and distributes the cryptographic keys required for secure communication.

(3) **Identity Authentication**. Vehicles and the AS mutually verify the legitimacy of their identities. A vehicle transmits communication data to the AS in encrypted form. The AS performs the required computations directly over the encrypted data, preserving data privacy and supporting functional utility.

(4) **Message Communication**. After identity authentication, the vehicle encrypts its data using the established cryptographic parameters and transmits it to the AS. The AS receives the communication message and decrypts the ciphertext or performs the required operations according to the application scenario. This process supports message confidentiality, integrity, and authenticity, enabling efficient and secure data utilization in VANETs.

C. Security Requirements

The proposed heterogeneous identity authentication protocol is required to satisfy the following security properties.

(1) **DoS Attack Resistance**. The protocol should resist service disruption caused by large volumes of forged requests targeting RSUs. Such requests may cause authentication failures, communication interruptions, and degradation of traffic safety.

(2) **Unforgeability**. Authentication messages must be cryptographically bound to users' private keys to prevent adversaries from impersonating legitimate entities or injecting forged data.

(3) **Replay Attack Prevention**. The protocol should detect and reject reused or replayed messages to prevent resource exhaustion, fraudulent operations, and information leakage.

(4) **Internal Attack Mitigation**. The system must resist attacks by malicious insiders, including compromised users or devices, to prevent unauthorized access, data manipulation, and system instability.

(5) **Man-in-the-Middle Defense**. The protocol should preserve end-to-end message integrity and prevent unauthorized interception, modification, or control of vehicular communications.

(6) **Privacy Preservation**. Sensitive vehicular information, including driving behavior and user identity, must be protected against privacy leakage and associated security risks.

(7) **Confidentiality**. Encryption mechanisms should ensure that sensitive data, including location information, is accessible only to authorized entities, such as the AS, and remains protected against adversarial interception and decryption.

IV. SECURE AND EFFICIENT HETEROGENEOUS IDENTITY AUTHENTICATION SCHEME

A. Formal Definition

The proposed heterogeneous identity authentication protocol consists of the following algorithms.

(1) **Initialization**. This phase initializes the IBC and PKI systems and generates their corresponding system parameters.

1) $(msk_1, Params_1) \leftarrow \text{CA-Setup}(1^\kappa, t)$. Given the security parameter κ and the threshold t , CA-Setup generates the master secret key msk_1 and the public parameters $Params_1$. The parameters $Params_1$ are treated as common inputs and are omitted hereafter for brevity.

2) $(msk_2, Params_2) \leftarrow \text{PKG-Setup}(1^\kappa, n)$. On input κ and the vector dimension n , PKG-Setup outputs the master secret key msk_2 and the public parameters $Params_2$. The parameters $Params_2$ are similarly omitted from subsequent descriptions.

(2) **Registration**. In this phase, the AS, RSUs, and vehicles complete their registration procedures with the CA and PKG and obtain the corresponding key material.

1) $rsk_j \leftarrow \text{RKGen}(RSU_j, msk_1)$. Executed by the CA, this algorithm takes the identity RSU_j and the master secret key msk_1 as inputs and outputs the private key rsk_j .

2) $sk_{j,i} \leftarrow \text{Partial-Key-Extract}(V_i, rsk_j)$. Using the identity of vehicle V_i , RSU_j generates the partial private key $sk_{j,i}$ and returns it to V_i .

3) $vs_{k_i} \leftarrow \text{VKGen}(sk_{1,i}, sk_{2,i}, \dots, sk_{t,i})$. Vehicle V_i derives its private key vs_{k_i} from the t received partial private keys $sk_{1,i}, sk_{2,i}, \dots, sk_{t,i}$.

4) $(ask_y, apk) \leftarrow \text{AKGen}(msk_2, ID_A, y)$. Given msk_2 , ID_A , and y , the PKG outputs the private key ask_y and the public key apk .

(3) **Authentication**. In this phase, the AS and vehicles perform mutual authentication.

1) $\delta \leftarrow \text{Prove}(ask_y, ID_A)$. The AS generates the authentication message δ using its private key ask_y and identity ID_A .

2) $0/1 \leftarrow \text{Verify}(\delta, apk, ID_A)$. Upon receiving the authentication message δ , the verifier executes the verification algorithm Verify using the AS's public key apk and identity ID_A . If δ is successfully validated, the verifier confirms that the message was generated by the legitimate AS and accepts its authenticity; otherwise, the message is rejected.

(4) **Communication**. After successful authentication, the AS and vehicles establish secure message communication.

1) $ct_i \leftarrow \text{Enc}(x_i)$. Each vehicle V_i takes its information x_i as input and outputs the corresponding ciphertext ct_i .

2) $CT_x \leftarrow \text{AggEnc}((ct_{x_i})_{i=1,\dots,d})$. This algorithm is executed by RSU_j . It takes the collected ciphertexts $\{ct_{x_i}\}_{i=1}^d$ as input and outputs the aggregated ciphertext CT_x .

3) $\langle x, y \rangle \leftarrow \text{Dec}(ask_y, CT_x)$. This algorithm is executed by the AS. Given ask_y and CT_x as inputs, it outputs the inner product $\langle x, y \rangle$.

B. Security Model

In the proposed heterogeneous identity authentication scheme, vehicles operate under the identity-based cryptographic system. The security analysis focuses on two properties of the corresponding cryptographic algorithms: selective indistinguishability under chosen-plaintext attacks (sIND-CPA) for encrypted communication messages and existential unforgeability under chosen-message attacks (EUF-CMA) for authentication messages. These properties characterize the confidentiality of the encrypted data and the unforgeability of authentication messages, respectively.

We consider two probabilistic polynomial-time (PPT) adversaries, $\mathcal{A}_1$ and $\mathcal{A}_2$, against sIND-CPA and EUF-CMA security, respectively.

1) *IND-CPA Security*: The following game formalizes the confidentiality of the communication messages of heterogeneous identity authentication scheme.

Game 1. This game is conducted between the challenger $\mathcal{C}$ and the adversary $\mathcal{A}_1$. The details are given as follows.

Initialization. $\mathcal{A}_1$ generates two equal-length vectors $\mathbf{x}_0^*$ and $\mathbf{x}_1^*$ of length n and sends them to $\mathcal{C}$.

Setup. $\mathcal{C}$ runs the algorithm $(msk_2, Params_2) \leftarrow \text{PKG-Setup}(1^\kappa, n)$ to generate the public parameters $Params_2$ and the master secret key msk_2 , and then sends $Params_2$ to $\mathcal{A}_1$.

Queries 1. $\mathcal{A}_1$ adaptively issues a polynomial number of secret key generation queries for a key vector $\mathbf{y}$, subject to the constraint that $\langle \mathbf{x}_0^* - \mathbf{x}_1^*, \mathbf{y} \rangle = 0$. For each admissible query, $\mathcal{C}$ runs the algorithm $(ask_{\mathbf{y}}, apk) \leftarrow \text{AKGen}(msk_2, ID_A, \mathbf{y})$ and returns $ask_{\mathbf{y}}$ to $\mathcal{A}_1$.

Challenge. $\mathcal{C}$ randomly selects a bit ω , generates the ciphertext $ct_i \leftarrow \text{Enc}(\mathbf{x}_{i,\omega})$, and sends it to $\mathcal{A}_1$.

Queries 2. This phase follows the same procedure and query constraint as the “Queries 1” phase.

Guess. $\mathcal{A}_1$ outputs a guess $\omega' \in \{0, 1\}$ for ω . The adversary wins the game if $\omega' = \omega$.

The advantage of $\mathcal{A}_1$ is defined as $\text{Adv}_{\mathcal{A}_1}^{\text{IND-CPA}}(\kappa) = \left| \Pr[\omega' = \omega] - \frac{1}{2} \right|$.

Definition 4 (IND-CPA Security): For the above PPT adversary $\mathcal{A}_1$, if the advantage $\text{Adv}_{\mathcal{A}_1}^{\text{IND-CPA}}(\kappa)$ is negligible, then the communication messages of heterogeneous identity authentication protocol achieves IND-CPA security.

2) *EUF-CMA Security*: The following game formalizes the unforgeability of authentication messages in the proposed heterogeneous identity authentication protocol.

Game 2. This game is conducted between the challenger $\mathcal{C}$ and the adversary $\mathcal{A}_2$. The game proceeds as follows.

Setup. $\mathcal{C}$ takes the security parameter κ and the vector length n as inputs and executes the algorithm $(msk_2, Params_2) \leftarrow \text{PKG-Setup}(1^\kappa, n)$ to generate the master secret key msk_2 and the system public parameters $Params_2$. The challenger sends $Params_2$ and keeps msk_2 secret.

Queries 1. $\mathcal{A}_2$ may issue the following queries.

(1) *Public Key Generation.* $\mathcal{A}_2$ submits a public key generation query associated with identity ID_i and vector $\mathbf{y}_i$. $\mathcal{C}$ executes the key generation algorithm $(ask_{\mathbf{y}_i}, apk_i) \leftarrow \text{AKGen}(msk_2, ID_i, \mathbf{y}_i)$ and returns the corresponding public key apk_i to $\mathcal{A}_2$.

(2) *Private Key Generation.* $\mathcal{A}_2$ submits a private key generation query with vector $\mathbf{y}_i$ and identity ID_i as inputs. $\mathcal{C}$ runs the key generation algorithm $(ask_{\mathbf{y}_i}, apk_i) \leftarrow \text{AKGen}(msk_2, ID_i, \mathbf{y}_i)$ and returns the corresponding private key $ask_{\mathbf{y}_i}$ to $\mathcal{A}_2$.

(3) *Authentication Message Generation.* Upon receiving an authentication message generation query for identity ID_i , $\mathcal{C}$ executes the authentication message generation algorithm $\delta_i \leftarrow \text{Prove}(ask_{\mathbf{y}_i}, ID_i)$ and returns the corresponding authentication message δ_i to $\mathcal{A}_2$. The required private key $ask_{\mathbf{y}_i}$ can be obtained by executing the private key generation algorithm.

Forgery. $\mathcal{A}_2$ generates a forged authentication message δ^* for the challenge identity ID^* . The adversary $\mathcal{A}_2$ wins the game if both of the following conditions hold: (1) the forged authentication message δ^* is valid; and (2) $\mathcal{A}_2$ has not issued a private key generation query related to the challenge identity ID^* .

The advantage of $\mathcal{A}_2$ is defined as $\text{Adv}_{\mathcal{A}_2}^{\text{EUF-CMA}}(\kappa) = \Pr[\mathcal{A}_2 \text{ wins}]$.

Definition 5 (EUF-CMA Security): For the above PPT adversary $\mathcal{A}_2$, if the advantage $\text{Adv}_{\mathcal{A}_2}^{\text{EUF-CMA}}(\kappa)$ is negligible, then the authentication messages of heterogeneous identity authentication protocol achieves EUF-CMA security.

V. OUR CONSTRUCTION OF HETEROGENEOUS IDENTITY AUTHENTICATION SCHEME

A. Concrete Construction

This section presents the concrete construction of our privacy protection protocol, which consists of four phases: system initialization, registration, mutual authentication, and message communication.

1) *System Initialization:* Given the security parameter κ , the vector length n , and the threshold η , the system selects two cyclic groups $\mathbb{G}_1, \mathbb{G}_2$ and a target group $\mathbb{G}_T$ of prime order q , together with a symmetric bilinear map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$. Let $g \in \mathbb{G}_1$ and $h \in \mathbb{G}_2$ be two generators.

(1) The CA executes the algorithm $\text{CA-Setup}(1^\kappa, \eta)$, randomly selects η values $s, a_1, \dots, a_{\eta-1} \in \mathbb{Z}_q^*$, and constructs a polynomial $f_1(x) = s + a_1x + a_2x^2 + \dots + a_{\eta-1}x^{\eta-1}$ of degree $\eta - 1$. It then computes $P = g^s$ as the system public key. The CA further selects three hash functions: $H : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$, and $H_5 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$. Finally, it generates the public parameters $Params_1 = \{g, e, P, H, H_1, H_5\}$ and the master secret key $msk_1 = s$.

(2) The PKG executes the algorithm $\text{PKG-Setup}(1^\kappa, n)$ and selects $\alpha_1, \alpha_2, \dots, \alpha_n, \beta \in \mathbb{Z}_q^*$. It computes $h_0 = h^\beta$, $g_{1,i} = g^{\alpha_i}$, and $h_{1,i} = h^{\alpha_i}$, where $i = 1, 2, \dots, n$. The PKG further selects three hash functions $H_2, H_3, H_4 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$. Finally, it establishes the public parameters as $Params_2 = \{g, g_1, g_2, \dots, g_n, e(g, h_0), H_2, H_3, H_4\}$ and sets the master secret key as $msk_2 = (h_0, h_1, h_2, \dots, h_n)$, which is kept confidential.

To ensure that all pairing operations in the subsequent authentication procedure are well defined, the construction adopts a symmetric bilinear setting in which the two source groups are instantiated as the same cyclic group.

2) *Registration*: RSUs from different domains can initiate registration requests to the CA, which assigns private keys according to their identity information. Each RSU can generate a corresponding partial private key for vehicles within its domain. The AS submits its vector $\mathbf{y}$ to the PKG as a registration request, and the PKG generates the corresponding key pair based on this request.

(1) The RSU sends its identity RSU_j to the CA. The CA executes the algorithm $RKGen(RSU_j, msk_1)$ to compute the private key $rsk_j = f_1(RSU_j)$ and returns it to RSU_j .

For the purpose of polynomial evaluation and Lagrange interpolation, each RSU identity is represented by a distinct nonzero element of the field $\mathbb{Z}_q^*$, and the same notation RSU_j is used for the corresponding field representation throughout the construction.

(2) Vehicle V_i randomly selects $k_i \in \mathbb{Z}_q^*$, computes $K_i = g^{k_i}$, and sends K_i together with its identity V_i as a request to RSU_j in its current domain. After receiving the request, RSU_j executes the algorithm $Partial-Key-Extract(V_i, rsk_j)$ to compute $c_i = H(K_i, V_i)$ and returns $sk_{j,i} = H_1(V_i)^{c_i \cdot rsk_j}$ to V_i . The partial private key of V_i is denoted by $sk_{j,i}$, and the identity of V_i serves as its public key. Due to vehicle mobility, V_i can request partial private keys from RSU_j in different domains. Once V_i receives $sk_{j,i}$ from at least η authorized RSUs, it can compute its private key as follows:

$$\begin{aligned} vsk_i &= H_1(V_i)^{k_i} \cdot \prod_{j=1}^{\eta} sk_{j,i}^{\prod_{k=1, k \neq j}^{\eta} \frac{RSU_k}{RSU_k - RSU_j}} \\ &= H_1(V_i)^{k_i} \cdot \prod_{j=1}^{\eta} H_1(V_i)^{c_i \cdot rsk_j \cdot \Delta_{j,\eta}(0)} \\ &= H_1(V_i)^{k_i} \cdot H_1(V_i)^{c_i \cdot \sum_{j=1}^{\eta} f_1(RSU_j) \cdot \Delta_{j,\eta}(0)} \\ &= H_1(V_i)^{k_i + c_i \cdot f_1(0)} = H_1(V_i)^{k_i + c_i \cdot s} \end{aligned} \quad (1)$$

(3) The AS sends the vector $\mathbf{y} = (y_1, y_2, \dots, y_n)$ to be evaluated and its identity ID_A to the PKG. The PKG executes the algorithm $AKGen(msk_2, ID_A, \mathbf{y})$ to generate keys for the AS. It randomly selects $r_1, r_2 \in \mathbb{Z}_q^*$ and computes $ask_1 = h_0 \cdot \prod_{i=1}^d h_{1,i}^{r_1 \cdot y_i}$, $ask_2 = h^{r_1}$, $ask_3 = e(g, h)^{r_1}$, and $ask_4 = r_2$. Finally, the PKG outputs the private key $ask_{\mathbf{y}} = (\mathbf{y}, ask_1, ask_2, ask_3, ask_4)$ and the public key $apk = g^{ask_4}$, and returns them to the AS.

3) *Mutual Authentication*: Before the AS communicates with multiple vehicles V_i , mutual identity authentication must be performed.

(1) Vehicle V_i randomly selects $t_i \in \mathbb{Z}_q^*$ and sends $T_i = H_1(V_i)^{t_i}$ and K_i to the AS.

(2) The AS randomly selects $t_1 \in \mathbb{Z}_q^*$ and computes $c_i = H(K_i, V_i)$ and $Q_i = e(T_i \cdot H_1(V_i)^{t_1}, K_i^{ask_4} \cdot P^{ask_4 \cdot c_i})$. It then computes $\gamma_i = H_2(Q_i)$ and randomly selects $\theta \in \mathbb{Z}_q^*$ to perform the following computation: $f_2(x) = \prod_{i=1}^d (x - \gamma_i) + \theta = x^d + b_{d-1}x^{d-1} + \dots + b_0$, where $b_i \in \mathbb{Z}_q^*$, and sets $B = \{b_0, b_1, \dots, b_{d-1}\}$. The AS further selects a random value $t_2 \in \mathbb{Z}_q^*$, computes $T_2 = g^{t_2}$ and $\sigma = h_3(t_2 + \theta) + h_4 \cdot ask_4$, where $h_3 = H_3(ID_A, T_2, \theta, B)$ and $h_4 = H_4(ID_A, T_2, \theta, B)$. Finally, it sends $\delta = \{\sigma, B, t_1, T_2\}$ to V_i .

(3) After receiving $\delta = \{\sigma, B, t_1, T_2\}$, V_i computes $Q'_i = e(vsk_i^{(t_i + t_1)}, apk)$ and $\gamma'_i = H_2(Q'_i)$. It then com-

putes $\theta' = f_2(\gamma'_i) = \gamma_i'^d + b_{d-1}\gamma_i'^{d-1} + \dots + b_0$, $h'_3 = H_3(ID_A, T_2, \theta', B)$, and $h'_4 = H_4(ID_A, T_2, \theta', B)$, where the function $f_2(x)$ can be reconstructed from the coefficient set $B = \{b_0, b_1, \dots, b_{d-1}\}$. If $g^\sigma = T_2^{h'_3} \cdot g^{h'_3 \cdot \theta'} \cdot apk^{h'_4}$, then the identity of the AS is verified by V_i . Next, V_i selects a random value $t'_i \in \mathbb{Z}_q^*$, computes $T'_i = g^{t'_i}$ and $\sigma_i = H_1(V_i)^{t'_i} \cdot vsk_i^{h'_5}$, where $h'_5 = H_5(V_i, T'_i, \theta')$ and $h'_6 = H_6(V_i, T'_i, \theta')$. Finally, V_i sends $\delta_i = \{\sigma_i, T'_i\}$ to the AS as the authentication response message.

(4) Upon receiving δ_i from V_i , the AS accepts the authentication only if $e(g, \sigma_i) = e(H_1(V_i), T'_i) \cdot e(H_1(V_i), K_i P^{c_i})^{h'_6}$, where $h'_5 = H_5(V_i, T'_i, \theta)$. Hence, the AS can verify the identity of V_i .

4) *Message Communication*: After all vehicles V_i within the domain of RSU_j complete authentication with the AS, each V_i encrypts its information $\mathbf{x}_i$ and sends the ciphertext to RSU_j . Then, RSU_j aggregates the ciphertexts generated by all vehicles in its domain and sends the aggregated ciphertext to the AS. Upon receiving the aggregated ciphertext, the AS decrypts it with its private key to obtain the required traffic assessment information.

(1) *Encryption*. Vehicle V_i randomly selects $z_i \in \mathbb{Z}_q^*$ and runs the algorithm $Enc(\mathbf{x}_i)$ to encrypt its information $\mathbf{x}_i$. The specific process of the encryption algorithm $Enc(\mathbf{x}_i)$ is as follows:

It computes $c_{i,1} = g^{z_i}$, $c_{i,2,j} = g^{x_{i,j}} g_{1,j}^{z_i}$, and $c_{i,3} = e(g, h_0)^{z_i}$. Let $\mathbf{c}_{i,2} = (c_{i,2,1}, c_{i,2,2}, \dots, c_{i,2,n})$. Therefore, the ciphertext generated by V_i is $ct_{\mathbf{x}_i} = (c_{i,1}, \mathbf{c}_{i,2}, c_{i,3})$, which is transmitted to RSU_j .

(2) *Ciphertext Aggregation*. After receiving the ciphertexts generated by the d authenticated vehicles within its domain, RSU_j executes the algorithm $AggEnc((ct_{\mathbf{x}_i})_{i=1,\dots,d})$. It computes $C_1 = \prod_{i=1}^d c_{i,1}$, $C_2 = (c_{1,2}, c_{2,2}, \dots, c_{d,2})$. It further computes $C_3 = \prod_{i=1}^d c_{i,3}$. Then, it sends the aggregated traffic information $CT_{\mathbf{x}} = (C_1, C_2, C_3)$ to the AS.

(3) *Functional Decryption*. Upon receiving $CT_{\mathbf{x}} = (C_1, C_2, C_3)$, the AS executes the algorithm $Dec(ask_{\mathbf{y}}, CT_{\mathbf{x}})$ using the functional key associated with $\mathbf{y} = (y_1, y_2, \dots, y_n)$.

The AS first computes $C_{\mathbf{y}} = \prod_{i=1}^d \prod_{j=1}^n c_{i,2,j}^{y_j}$. That is, using $c_{i,2,j} = g^{x_{i,j}} g_{1,j}^{z_i}$ and $g_{1,j} = g^{\alpha_j}$, it follows that $C_{\mathbf{y}} = g^{\sum_{i=1}^d \langle \mathbf{x}_i, \mathbf{y} \rangle + \sum_{i=1}^d \sum_{j=1}^n \alpha_j z_i y_j}$. After that, AS computes $D = e(C_1, ask_1)^{-1} \cdot e(C_{\mathbf{y}}, ask_2) \cdot C_3$.

Finally, the AS obtains the aggregated traffic assessment result as $\log_{ask_3} D$. According to the linearity of the inner product, the recovered result can equivalently be expressed as $\log_{ask_3} D$.

B. Correctness

Functional Decryption Correctness. According to the encryption algorithm, for each vehicle V_i and each vector dimension j , we have $g^{x_{i,j}} g_{1,j}^{z_i} = g^{x_{i,j} + \alpha_j z_i}$. Therefore, we can obtain

$$\begin{aligned} C_{\mathbf{y}} &= \prod_{i=1}^d \prod_{j=1}^n c_{i,2,j}^{y_j} = \prod_{i=1}^d \prod_{j=1}^n (g^{x_{i,j} + \alpha_j z_i})^{y_j} \\ &= g^{\sum_{i=1}^d \sum_{j=1}^n x_{i,j} y_j} \cdot g^{\sum_{i=1}^d \sum_{j=1}^n \alpha_j z_i y_j} \\ &= g^{\sum_{i=1}^d \langle \mathbf{x}_i, \mathbf{y} \rangle} \cdot g^{\sum_{i=1}^d z_i \sum_{j=1}^n \alpha_j y_j} \end{aligned} \quad (2)$$

Meanwhile, $C_1 = \prod_{i=1}^d g^{z_i}$ and $C_3 = \prod_{i=1}^d e(g, h_0)^{z_i}$. The functional key satisfies $ask_1 = h_0 \cdot \prod_{j=1}^n h_{1,j}^{r_1 y_j}$, where $h_{1,j} = h^{\alpha_j}$. Hence, we have $ask_1 = h_0 \cdot h^{r_1 \sum_{j=1}^n \alpha_j y_j}$. Then, we can obtain

$$\begin{aligned} D &= e(C_1, ask_1)^{-1} \cdot e(C_y, ask_2) \cdot C_3 \\ &= e\left(\prod_{i=1}^d g^{z_i}, h_0\right)^{-1} \cdot e\left(\prod_{i=1}^d g^{z_i}, h^{r_1 \sum_{j=1}^n \alpha_j y_j}\right)^{-1} \\ &\quad \cdot e\left(g^{\sum_{i=1}^d \langle x_i, y \rangle}, h^{r_1}\right) \\ &\quad \cdot e\left(g^{\sum_{i=1}^d z_i \sum_{j=1}^n \alpha_j y_j}, h^{r_1}\right) \cdot \prod_{i=1}^d e(g, h_0)^{z_i} \\ &= e\left(g^{\sum_{i=1}^d \langle x_i, y \rangle}, h^{r_1}\right) \\ &= e(g, h)^{r_1 \sum_{i=1}^d \langle x_i, y \rangle} = ask_3^{\sum_{i=1}^d \langle x_i, y \rangle} \end{aligned} \quad (3)$$

Identity Authentication Correctness. During mutual authentication, the following derivation shows that the AS and a legitimate vehicle obtain the same pairing value.

$$\begin{aligned} Q_i &= e\left(T_i \cdot H_1(V_i)^{t_1}, K_i^{ask_4} \cdot P^{ask_4 \cdot c_i}\right) \\ &= e\left(H_1(V_i)^{t_i+t_1}, g^{ask_4 \cdot (k_i+s \cdot c_i)}\right) \\ &= e\left(H_1(V_i)^{(t_i+t_1) \cdot (k_i+s \cdot c_i)}, g^{ask_4}\right) \\ &= e\left(vsk_i^{(t_i+t_1)}, apk\right) = Q'_i \end{aligned} \quad (4)$$

Therefore, a legitimate vehicle obtains $Q'_i = Q_i$ and consequently $\gamma'_i = \gamma_i$. According to the construction of $f_2(x)$, evaluating the polynomial at γ'_i eliminates one factor of the product and gives $\theta' = \theta$. The hash inputs at the AS and the vehicle are therefore identical, implying $h'_3 = h_3$ and $h'_4 = h_4$. Hence, the AS authentication equation is satisfied.

Similarly, since $\theta' = \theta$, the vehicle and the AS obtain identical values for the corresponding hash outputs in the response verification step. Therefore, the stated verification equation for σ_i holds for an honestly generated authentication response.

$$\begin{aligned} T_2^{h'_3} \cdot g^{h'_3 \cdot \theta'} \cdot apk^{h'_4} &= g^{t_2 \cdot h'_3} \cdot g^{h'_3 \cdot \theta'} \cdot g^{h'_4 \cdot ask_4} \\ &= g^{h'_3 \cdot (t_2 + \theta') + h'_4 \cdot ask_4} \\ &= g^{h_3 \cdot (t_2 + \theta) + h_4 \cdot ask_4} = g^\sigma \end{aligned} \quad (5)$$

According to the vehicle authentication response $\sigma_i = H_1(V_i)^{t'_i} \cdot vsk_i^{h_5}$, we have

$$\begin{aligned} e(\sigma_i, g) &= e\left(H_1(V_i)^{t'_i} \cdot vsk_i^{h_5}, g\right) \\ &= e(H_1(V_i), g^{t'_i}) \cdot e(vsk_i, g)^{h_5} \\ &= e(H_1(V_i), T'_i) \cdot e\left(H_1(V_i)^{k_i+s \cdot c_i}, g\right)^{h_5} \\ &= e(H_1(V_i), T'_i) \cdot e\left(H_1(V_i), g^{k_i+s \cdot c_i}\right)^{h_5} \\ &= e(H_1(V_i), T'_i) \cdot e\left(H_1(V_i), K_i P^{c_i}\right)^{h_5} \\ &= e(H_1(V_i), T'_i) \cdot e\left(H_1(V_i), K_i P^{c_i}\right)^{h'_5}. \end{aligned} \quad (6)$$

Hence, an honestly generated vehicle authentication response satisfies the verification equation. Since the response explicitly depends on vsk_i , knowledge of the common value

θ alone is insufficient to generate a valid response for another vehicle.

VI. PROOF, ANALYSIS AND SIMULATION OF SECURITY

In this section, we analyze the security of the above heterogeneous identity authentication scheme from three perspectives: security proof, security analysis, and security simulation.

A. Security Proof

1) Selective IND-CPA Security: The confidentiality of communication messages ensures the security of the heterogeneous identity authentication protocol. Therefore, in this section, we will prove the selective IND-CPA security of the encryption algorithm Enc. To substantiate the selective IND-CPA security, we construct a sequence of hybrid games. These games gradually transform the encryption of an initial message x_0 into the encryption of an alternative message x_1 . The security of the proposed scheme is established through a fundamental theorem, which is proven under the XDH assumption.

Theorem 1: Under the XDH assumption, the proposed IPFE-based encryption scheme achieves selective IND-CPA security against any PPT adversary.

Proof 1: We prove the selective IND-CPA security of the proposed IPFE encryption scheme through a sequence of hybrid games. Before the system setup, the adversary $\mathcal{A}_1$ selects two challenge vectors $x_0 = (x_{0,1}, \dots, x_{0,n})$ and $x_1 = (x_{1,1}, \dots, x_{1,n})$ and sends them to the challenger $\mathcal{C}$. Define the difference vector as $\Delta = x_1 - x_0 = (\Delta_1, \dots, \Delta_n)$. For every admissible functional-key query y , the security-game restriction implies $\langle \Delta, y \rangle = 0$.

According to the revised encryption algorithm, a ciphertext of x_i is represented as $ct_{x_i} = (c_{i,1}, c_{i,2}, c_{i,3}) \in \mathbb{G}_1 \times \mathbb{G}_1^n \times \mathbb{G}_T$, where $c_{i,2} = (c_{i,2,1}, \dots, c_{i,2,n})$. Given encryption randomness $z_i \in \mathbb{Z}_q^*$, the ciphertext components are $c_{i,1} = g_i^{z_i}$, $c_{i,2} = g^{x_{i,j}} g_{1,j}^{z_i}$ for $j = 1, \dots, n$, and $c_{i,3} = e(g, h_0)^{z_i}$.

We consider the following sequence of hybrid games.

Game $\mathcal{G}_{0,0}$. This is the original left challenge game, in which the challenger encrypts x_0 .

Initialization. The adversary $\mathcal{A}_1$ selects the challenge vectors x_0 and x_1 and sends them to $\mathcal{C}$.

Setup. The challenger randomly selects $\beta \in \mathbb{Z}_q^*$ and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_q^n$. For each $j = 1, \dots, n$, it computes $g_{1,j} = g^{\alpha_j}$. It further computes $h_0 = h^\beta$ and sets the public parameters relevant to the encryption scheme as $Params = (g, \{g_{1,j}\}_{j=1}^n, e(g, h_0))$. The corresponding secret information is retained by $\mathcal{C}$ for functional-key generation.

Queries 1. The adversary $\mathcal{A}_1$ adaptively issues a polynomial number of key generation queries for vectors $y = (y_1, \dots, y_n)$ satisfying $\langle x_{i,0}, y \rangle = \langle x_{i,1}, y \rangle$. For each admissible query, $\mathcal{C}$ randomly selects $r_1 \in \mathbb{Z}_q^*$ and computes $ask_1 = h_0 \cdot h^{r_1 \sum_{j=1}^n \alpha_j y_j}$, $ask_2 = h^{r_1}$, and $ask_3 = e(g, h)^{r_1}$. The functional key is denoted by $ask_y = (y, ask_1, ask_2, ask_3)$ and is returned to $\mathcal{A}_1$.

The authentication private key used by the AS is generated independently of the IPFE functional key and therefore does not participate in the confidentiality game.

Challenge. The challenger randomly selects $z \in \mathbb{Z}_q^*$ and computes $c_1^* = g^z$, $c_{2,j}^* = g^{x_{0,j}} g_{1,j}^z$ for $j = 1, \dots, n$, and $c_3^* = e(g, h_0)^z$. It then sends the challenge ciphertext $ct^* = (c_1^*, c_2^*, c_3^*)$ to $\mathcal{A}_1$, where $c_2^* = (c_{2,1}^*, \dots, c_{2,n}^*)$.

Queries 2. This phase is identical to the “Queries 1” phase. Each queried vector $\mathbf{y}$ must satisfy the same inner-product restriction.

Guess. The adversary outputs its guess based on the received challenge ciphertext.

Game $\mathcal{G}_{0,1}$. This game is identical to $\mathcal{G}_{0,0}$ except that the challenger encrypts $\mathbf{x}_0 + t\Delta$, where $t \leftarrow \mathbb{Z}_q$ is hidden from $\mathcal{A}_1$. The components c_1^* and c_3^* remain unchanged, while the j -th component of $c_{2,j}^*$ becomes $c_{2,j}^* = g^{x_{0,j} + t\Delta_j} g_{1,j}^z$.

We show that $\mathcal{G}_{0,0}$ and $\mathcal{G}_{0,1}$ are computationally indistinguishable under the XDH assumption. Suppose that a PPT adversary can distinguish these two games with non-negligible advantage. We construct an algorithm $\mathcal{B}_0$ that distinguishes an XDH instance.

Let $(g, X, Y, Z) = (g, g^a, g^b, Z)$ denote an XDH challenge tuple, where either $Z = g^{ab}$ or $Z = g^c$ for a uniformly random $c \in \mathbb{Z}_q$. Algorithm $\mathcal{B}_0$ randomly selects $\tilde{\alpha} = (\tilde{\alpha}_1, \dots, \tilde{\alpha}_n) \in \mathbb{Z}_q^n$ and defines $g_{1,j} = g^{\tilde{\alpha}_j} X^{\Delta_j}$ for each $j = 1, \dots, n$. The corresponding implicit exponent is $\alpha_j = \tilde{\alpha}_j + a\Delta_j$.

Although $\mathcal{B}_0$ does not know a , it can answer every admissible functional-key query. For a queried vector $\mathbf{y}$, we have

$$\begin{aligned} \sum_{j=1}^n \alpha_j y_j &= \sum_{j=1}^n (\tilde{\alpha}_j + a\Delta_j) y_j \\ &= \sum_{j=1}^n \tilde{\alpha}_j y_j + a \sum_{j=1}^n \Delta_j y_j = \sum_{j=1}^n \tilde{\alpha}_j y_j, \end{aligned} \quad (7)$$

where the final equality follows from $\langle \Delta, \mathbf{y} \rangle = 0$. Therefore, $\mathcal{B}_0$ can generate $ask_1 = h_0 \cdot h^{r_1 \sum_{j=1}^n \tilde{\alpha}_j y_j}$ without knowing a . The remaining functional-key components are generated normally.

For the challenge ciphertext, $\mathcal{B}_0$ sets $c_1^* = Y$, computes $c_{2,j}^* = g^{x_{0,j}} Y^{\tilde{\alpha}_j} Z^{\Delta_j}$ for each $j = 1, \dots, n$, and sets $c_3^* = e(Y, h_0)$.

If $Z = g^{ab}$, then we have

$$\begin{aligned} c_{2,j}^* &= g^{x_{0,j}} g^{b\tilde{\alpha}_j} g^{ab\Delta_j} \\ &= g^{x_{0,j}} \left(g^{\tilde{\alpha}_j + a\Delta_j} \right)^b = g^{x_{0,j}} g_{1,j}^b. \end{aligned} \quad (8)$$

Hence, the challenge ciphertext is distributed exactly as in $\mathcal{G}_{0,0}$ with encryption randomness $z = b$.

If $Z = g^c$ is random, define $t = c - ab$. Since c is uniformly distributed over $\mathbb{Z}_q$, the value t is also uniformly distributed over $\mathbb{Z}_q$. In this case,

$$\begin{aligned} c_{2,j}^* &= g^{x_{0,j}} g^{b\tilde{\alpha}_j} g^{c\Delta_j} \\ &= g^{x_{0,j} + (c-ab)\Delta_j} g^{b(\tilde{\alpha}_j + a\Delta_j)} = g^{x_{0,j} + t\Delta_j} g_{1,j}^b. \end{aligned} \quad (9)$$

Therefore, the challenge ciphertext is distributed exactly as in $\mathcal{G}_{0,1}$. Any non-negligible advantage in distinguishing $\mathcal{G}_{0,0}$ from $\mathcal{G}_{0,1}$ would consequently yield a non-negligible advantage against the XDH assumption. Thus, $\mathcal{G}_{0,0} \approx_c \mathcal{G}_{0,1}$.

Game $\mathcal{G}_{1,1}$. In this game, the challenger encrypts $\mathbf{x}_1 + (t-1)\Delta$. By the definition $\Delta = \mathbf{x}_1 - \mathbf{x}_0$, we obtain

$$\begin{aligned} \mathbf{x}_1 + (t-1)\Delta &= \mathbf{x}_1 + (t-1)(\mathbf{x}_1 - \mathbf{x}_0) \\ &= \mathbf{x}_0 + t(\mathbf{x}_1 - \mathbf{x}_0) = \mathbf{x}_0 + t\Delta. \end{aligned} \quad (10)$$

Therefore, $\mathcal{G}_{0,1}$ and $\mathcal{G}_{1,1}$ encrypt exactly the same vector for the same value of t . All other parameters and ciphertext-generation procedures are identical. Hence, $\mathcal{G}_{0,1} \equiv \mathcal{G}_{1,1}$, and no computational assumption is required for this transition.

Game $\mathcal{G}_{1,0}$. This is the original right challenge game, in which the challenger encrypts $\mathbf{x}_1$. We establish the computational indistinguishability between $\mathcal{G}_{1,1}$ and $\mathcal{G}_{1,0}$ through a symmetric reduction to the XDH assumption.

Given an XDH challenge tuple $(g, X, Y, Z) = (g, g^a, g^b, Z)$, the algorithm $\mathcal{B}_1$ randomly selects $\tilde{\alpha} = (\tilde{\alpha}_1, \dots, \tilde{\alpha}_n) \in \mathbb{Z}_q^n$ and defines $g_{1,j} = g^{\tilde{\alpha}_j} X^{-\Delta_j}$. The implicit exponent is $\alpha_j = \tilde{\alpha}_j - a\Delta_j$.

For every admissible query $\mathbf{y}$, we have

$$a \sum_{j=1}^n \Delta_j y_j = \sum_{k=1}^n \tilde{\alpha}_k y_k. \quad (11)$$

Thus, $\mathcal{B}_1$ can answer all admissible functional-key queries without knowing a .

For the challenge ciphertext, $\mathcal{B}_1$ sets $c_1^* = Y$, computes $c_{2,j}^* = g^{x_{1,j}} Y^{\tilde{\alpha}_j} Z^{-\Delta_j}$ for each $j = 1, \dots, n$, and sets $c_3^* = e(Y, h_0)$.

If $Z = g^{ab}$, then we obtain

$$\begin{aligned} c_{2,j}^* &= g^{x_{1,j}} g^{b\tilde{\alpha}_j} g^{-ab\Delta_j} \\ &= g^{x_{1,j}} \left(g^{\tilde{\alpha}_j - a\Delta_j} \right)^b = g^{x_{1,j}} g_{1,j}^b. \end{aligned} \quad (12)$$

Hence, the resulting challenge ciphertext is distributed exactly as in $\mathcal{G}_{1,0}$.

If $Z = g^c$ is random, define $t = 1 + ab - c$, which gives $t - 1 = ab - c$. We then obtain

$$\begin{aligned} c_{2,j}^* &= g^{x_{1,j}} g^{b\tilde{\alpha}_j} g^{-c\Delta_j} \\ &= g^{x_{1,j} + (ab-c)\Delta_j} g^{b(\tilde{\alpha}_j - a\Delta_j)} = g^{x_{1,j} + (t-1)\Delta_j} g_{1,j}^b. \end{aligned} \quad (13)$$

Therefore, the challenge ciphertext is distributed exactly as in $\mathcal{G}_{1,1}$. It follows that $\mathcal{G}_{1,1} \approx_c \mathcal{G}_{1,0}$ under the XDH assumption.

For $b = \{0, 1\}$, let $P_{b,s} = \Pr[\mathcal{A}_1 \text{ outputs } 1 \text{ in } \mathcal{G}_{b,s}]$. By the triangle inequality, we can obtain

$$\begin{aligned} |P_{0,0} - P_{1,0}| &\leq |P_{0,0} - P_{0,1}| + |P_{0,1} - P_{1,1}| \\ &\quad + |P_{1,1} - P_{1,0}|. \end{aligned} \quad (14)$$

The first and third terms are negligible under the XDH assumption. Since $\mathcal{G}_{0,1}$ and $\mathcal{G}_{1,1}$ are identically distributed, $|P_{0,1} - P_{1,1}| = 0$. Consequently, $|P_{0,0} - P_{1,0}|$ is negligible.

Therefore, no PPT adversary can distinguish an encryption of $\mathbf{x}_0$ from an encryption of $\mathbf{x}_1$ with non-negligible advantage, provided that all queried functional vectors $\mathbf{y}$ satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$. Hence, the revised IPFE-based encryption scheme achieves selective sIND-CPA security under the XDH assumption.

2) *Unforgeability*: The unforgeability of the identity authentication request message ensures the credibility of the identity authentication result. Below, we will prove the unforgeability of the authentication request message.

Theorem 2: Assume that the DL and the CDH assumptions, in the random oracle model, the authentication message and the authentication response of the proposed heterogeneous identity authentication protocol are existentially unforgeable under chosen-message attacks.

Proof 2: The mutual authentication procedure contains two authentication directions with different algebraic structures. We therefore analyze the unforgeability of the AS authentication message and that of the vehicle authentication response separately.

3) *Unforgeability of the AS Authentication Message*: We first prove that forging a valid AS authentication message implies an efficient solution to the DL problem. Suppose that a PPT adversary $\mathcal{A}_2$ can generate a valid forgery for the target AS with a non-negligible probability ε_A . We construct a challenger $\mathcal{C}_A$ that uses $\mathcal{A}_2$ to solve the DL problem.

Setup. The challenger $\mathcal{C}_A$ receives a DL instance (g, Y) , where $Y = g^b$ and $b \in \mathbb{Z}_q^*$ is unknown. The objective of $\mathcal{C}_A$ is to recover b . It embeds the DL challenge into the public key of the target AS by setting $apk^* = Y$ and implicitly setting $ask_4^* = b$.

All remaining system parameters are generated honestly according to the system initialization algorithm. In particular, the CA and PKG parameters that are independent of ask_4^* can be generated normally. The challenger initializes the random-oracle lists L_3 and L_4 for H_3 and H_4 , respectively.

The adversary is not allowed to obtain the long-term authentication private key ask_4^* of the target AS. Otherwise, the unforgeability experiment would become trivial. Private-key queries for non-target entities can be answered honestly.

H_3 -Oracle. When $\mathcal{A}_2$ submits an H_3 oracle query on (ID_A, T_2, θ, B) , $\mathcal{C}_A$ first checks L_3 . If a corresponding tuple has already been recorded, the previously generated value is returned. Otherwise, $\mathcal{C}_A$ randomly selects $h_3 \in \mathbb{Z}_q^*$, stores $(ID_A, T_2, \theta, B, h_3)$ in L_3 , and returns h_3 .

H_4 -Oracle. The H_4 oracle is simulated similarly. For a new query (ID_A, T_2, θ, B) , $\mathcal{C}_A$ randomly selects $h_4 \in \mathbb{Z}_q^*$, records $(ID_A, T_2, \theta, B, h_4)$ in L_4 , and returns h_4 .

Authentication Message Generation. When $\mathcal{A}_2$ requests an authentication message for the target AS, $\mathcal{C}_A$ must simulate a valid message without knowing $ask_4^* = b$.

The protocol values associated with the vehicles, including t_1 , θ , and the coefficient set B , can be generated according to the protocol. In particular, because $\mathcal{C}_A$ generates the vehicle-side parameters itself, it knows $k_i + sc_i$ and can compute the value required in the pairing through $apk^{*, k_i + sc_i}$ without knowing b . Hence, the polynomial-generation procedure can be simulated consistently with the real protocol.

The challenger then randomly selects $\sigma, h_3, h_4 \in \mathbb{Z}_q^*$ and computes $T_2 = (g^\sigma g^{-h_3\theta} (apk^*)^{-h_4})^{h_3^{-1}}$. It programs the random oracles such that $h_3 = H_3(ID_A, T_2, \theta, B)$ and $h_4 = H_4(ID_A, T_2, \theta, B)$, and returns $\delta = (\sigma, B, t_1, T_2)$ to $\mathcal{A}_2$. If one of these oracle inputs has already been defined inconsis-

tently, $\mathcal{C}_A$ repeats the simulation with fresh randomness. The probability of such a collision is negligible.

The simulated authentication message satisfies $T_2^{h_3} = g^{-\sigma} g^{h_3\theta} (apk^*)^{h_4}$. Moreover, there exists an implicit value $t_2 \in \mathbb{Z}_q$ satisfying $T_2 = g^{t_2}$ and $\sigma = h_3(t_2 + \theta) + h_4b$. Since σ is uniformly random and $h_3 \neq 0$, the implied value t_2 is uniformly distributed. Therefore, the simulated authentication message has the same distribution as an honestly generated message.

Forgery. After a polynomial number of oracle and authentication-message queries, $\mathcal{A}_2$ outputs a forged authentication message $\delta^* = \sigma^*, B^*, t_1^*, T_2^*$ for the target AS. Let θ^* denote the value reconstructed by an honest target vehicle from B^* during the verification procedure. The forgery is considered successful only if it corresponds to a fresh authentication transcript and satisfies $(T_2^*)^{h_3^*} = g^{h_3^*\theta^*} (apk^*)^{h_4^*}$, where $h_3^* = H_3(ID_A, T_2^*, \theta^*, B^*)$ and $h_4^* = H_4(ID_A, T_2^*, \theta^*, B^*)$.

Except with probability at most $1/(q-1)$, a successful adversary must have queried H_4 on $(ID_A, T_2^*, \theta^*, B^*)$. Otherwise, it would have to correctly guess the random-oracle output h_4^* .

According to the Forking Lemma [34], $\mathcal{C}_A$ rewinds $\mathcal{A}_2$ with the same random tape and identical oracle responses before the critical H_4 query. At this query, $\mathcal{C}_A$ returns a fresh value $\hat{h}_4^* \neq h_4^*$ while keeping the output of H_3 unchanged. With non-negligible probability, $\mathcal{A}_2$ generates a second valid forgery $\hat{\delta}^* = (\hat{\sigma}^*, B^*, t_1^*, T_2^*)$ satisfying $(T_2^*)^{h_3^*} = g^{h_3^*\theta^*} (apk^*)^{\hat{h}_4^*}$.

Dividing the two valid verification equations gives $(apk^*)^{h_4^* - \hat{h}_4^*} = g^{b(h_4^* - \hat{h}_4^*)}$. Since $h_4^* \neq \hat{h}_4^*$, $\mathcal{C}_A$ obtains $b = (\sigma^* - \hat{\sigma}^*)(h_4^* - \hat{h}_4^*)^{-1} \bmod q$.

Thus, $\mathcal{C}_A$ solves the DL problem. According to the Forking Lemma, if ε_A is non-negligible and the number of H_4 queries is polynomial in κ , then the probability that $\mathcal{C}_A$ obtains the two required valid forgeries is also non-negligible. This contradicts the assumed hardness of the DL problem. Therefore, the AS authentication message is existentially unforgeable under chosen-message attacks.

4) *Unforgeability of the Vehicle Authentication Response*: We next prove the unforgeability of the vehicle authentication response. According to the proposed construction, vehicle V_i generates $\sigma_i = H_1(V_i)^{t'_i} \cdot vsk_i^{h_5}$, where $T_i' = g^{t'_i}$, $h_5 = H_5(V_i, T_i', \theta')$, and $vsk_i = H_1(V_i)^{k_i + sc_i}$. Since $K_i = g^{k_i}$, $P = g^s$, and $c_i = H(K_i, V_i)$, the public relation $K_i P^{c_i} = g^{k_i} \cdot g^{sc_i} = g^{k_i + sc_i}$ holds. After computing $h_5' = H_5(V_i, T_i', \theta)$, the AS accepts the response only if $e(g, \sigma_i) = e(H_1(V_i), T_i') \cdot e(H_1(V_i), K_i P^{c_i})^{h_5'}$.

For an honestly generated authentication response, $\theta' = \theta$ and hence $h_5' = h_5$. The verification equation is correct because

$$\begin{aligned} e(g, \sigma_i) &= e(g, H_1(V_i)^{t'_i} \cdot vsk_i^{h_5}) \\ &= e(g, H_1(V_i))^{t'_i} \cdot e(g, vsk_i)^{h_5} \\ &= e(H_1(V_i), g^{t'_i}) \cdot e(H_1(V_i), g^{k_i + sc_i})^{h_5} \\ &= e(H_1(V_i), T_i') \cdot e(H_1(V_i), K_i P^{c_i})^{h_5} \end{aligned} \quad (15)$$

Here, the equalities $e(g, H_1(V_i)) = e(H_1(V_i), g)$ and $e(g, vsk_i) = e(vsk_i, g)$ follow from the symmetric bilinear setting adopted by the construction.

Suppose that a PPT adversary $\mathcal{A}_2$ can forge a fresh valid authentication response for a target vehicle V^* with a non-negligible probability ε_V . We construct a PPT challenger $\mathcal{C}_V$ that uses $\mathcal{A}_2$ to solve the CDH problem. We first consider the case in which the target vehicle V^* is fixed in the unforgeability experiment. If the experiment permits adaptive target selection, $\mathcal{C}_V$ can guess the target vehicle among the polynomially many vehicle identities considered by $\mathcal{A}_2$, which introduces only a polynomial loss in the reduction probability.

Setup. The challenger $\mathcal{C}_V$ receives a CDH instance (g, U, X) , where $U = g^a$ and $X = g^b$ for unknown $a, b \in \mathbb{Z}_q^*$, and its objective is to compute g^{ab} . The challenger honestly generates the CA master secret s and sets $P = g^s$. It programs the random oracle H_1 for the target vehicle as $H_1(V^*) = U = g^a$. The challenger then randomly selects $c^* \in \mathbb{Z}_q^*$ and defines $K^* = XP^{-c^*}$. If $K^* = 1_{\mathbb{G}}$, which occurs only with negligible probability, the challenger resamples c^* . It then programs the random oracle H such that $H(K^*, V^*) = c^*$.

By construction, $K^*P^{c^*} = XP^{-c^*}P^{c^*} = X = g^b$. Since $K^* \in \mathbb{G}$ and g generates $\mathbb{G}$, there exists an implicit exponent $k^* \in \mathbb{Z}_q^*$ satisfying $K^* = g^{k^*}$. Therefore, $g^{k^*+sc^*} = K^*P^{c^*} = g^b$, which implies $k^* + sc^* = b \pmod{q}$. Consequently, the private key implicitly associated with the target vehicle is $usk^* = H_1(V^*)^{k^*+sc^*} = U^b = (g^a)^b = g^{ab}$. Thus, computing usk^* is exactly equivalent to solving the given CDH instance.

The challenger can simulate partial private keys for the target vehicle without knowing usk^* . For an RSU with private share $rskj$, it computes $sk_j = U^{c^*rskj} = H_1(V^*)^{c^*rskj}$, which has the same form and distribution as a partial private key generated by the real protocol. As required by the unforgeability experiment, $\mathcal{A}_2$ is not allowed to obtain the complete private key usk^* of the target vehicle; otherwise, impersonation of V^* would be trivial.

H_5 -Oracle. The challenger maintains a list L_5 for the random oracle H_5 . Whenever $\mathcal{A}_2$ submits a new query (V_i, T'_i, θ) , $\mathcal{C}_V$ independently samples $h_5 \leftarrow \mathbb{Z}_q^*$, stores (V_i, T'_i, θ, h_5) in L_5 , and returns h_5 . If the same query has already appeared in L_5 , the challenger returns the previously assigned value, thereby providing a consistent random-oracle simulation.

Authentication Response Generation. For every non-target vehicle, $\mathcal{C}_V$ generates authentication responses honestly because it knows the corresponding private key. For an authentication-response query involving the target vehicle V^* and a session value θ , the challenger must generate a correctly distributed response without knowing $usk^* = U^b$.

To simulate such a response, $\mathcal{C}_V$ independently samples $h_5, u \leftarrow \mathbb{Z}_q^*$ and computes $\sigma^* = U^u$ and $T'^* = g^u X^{-h_5}$. If $T'^* = 1_{\mathbb{G}}$, the challenger resamples u . It also resamples whenever (V^*, T'^*, θ) has already been assigned a different value by the random oracle. Since the number of oracle queries is polynomial and T'^* ranges over a prime-order group of exponential size, the probability of such a collision is negligible. The challenger then programs $H_5(V^*, T'^*, \theta) = h_5$ and returns $\delta = \sigma^*, T'^*$.

The simulated response satisfies the real verification equa-

tion. Since $H_1(V^*) = U$ and $K^*P^{c^*} = X$, we obtain

$$\begin{aligned} & e(H_1(V^*), T'^*) \cdot e(H_1(V^*), K^*P^{c^*})^{h_5} \\ &= e(U, g^u X^{-h_5}) \cdot e(U, X)^{h_5} \\ &= e(U, g^u) \cdot e(U, X)^{-h_5} \cdot e(U, X)^{h_5} \\ &= e(U, g^u) = e(g, \sigma^*) \end{aligned} \quad (16)$$

Therefore, we have

$$e(g, \sigma^*) = e(H_1(V^*), T'^*) \cdot e(H_1(V^*), K^*P^{c^*})^{h_5},$$

which is exactly the verification equation of the proposed protocol. The simulated response also has the same distribution as an honestly generated response. Define the implicit value $t'^* = u - bh_5 \pmod{q}$. Since $X = g^b$, we have $T'^* = g^u X^{-h_5} = g^u (g^b)^{-h_5} = g^{u-bh_5} = g^{t'^*}$. Moreover, $\sigma^* = U^u = U^{t'^*+bh_5} = U^{t'^*} \cdot (U^b)^{h_5} = H_1(V^*)^{t'^*} \cdot (usk^*)^{h_5}$. Hence, δ^* has exactly the algebraic form of an honestly generated vehicle authentication response. Because u is uniformly distributed over $\mathbb{Z}_q$ for every fixed h_5 , the value $t'^* = u - bh_5$ is uniformly distributed over $\mathbb{Z}_q$; after rejecting the case $t'^* = 0$, it is uniformly distributed over $\mathbb{Z}_q^*$. Therefore, except for the negligible random-oracle collision event described above, the simulated response is distributed identically to an honestly generated response.

Forgery. After making a polynomial number of oracle and authentication-response queries, $\mathcal{A}_2$ outputs a fresh forged response $\delta_1^* = \sigma_1^*, T'^*$ for the target vehicle V^* and a session value θ^* . The forgery is required to satisfy the freshness condition of the EUF-CMA experiment and therefore cannot simply be a previously returned target authentication response. Let $h_5^* = H_5(V^*, T'^*, \theta^*)$. Since the forged response is valid, it must satisfy $e(g, \sigma_1^*) = e(H_1(V^*), T'^*) \cdot e(H_1(V^*), K^*P^{c^*})^{h_5^*} = e(U, T'^*) \cdot e(U, X)^{h_5^*}$.

Except with probability at most $1/(q-1)$, a successful adversary must have queried H_5 on (V^*, T'^*, θ^*) before producing the forgery. Otherwise, the value h_5^* remains uniformly random in $\mathbb{Z}_q^*$ from the adversary's perspective, and the adversary would have to guess the unique random-oracle value required by the verification equation.

According to the Forking Lemma [34], $\mathcal{C}_V$ rewinds $\mathcal{A}_2$ to the point immediately before this critical H_5 query while preserving the adversary's random tape and all random-oracle responses preceding the query. In the second execution, $\mathcal{C}_V$ answers the same query with a fresh value $\hat{h}_5^* \neq h_5^*$. With non-negligible probability, $\mathcal{A}_2$ outputs a second valid forgery $\delta_2^* = \sigma_2^*, T'^*$ for the same tuple (V^*, T'^*, θ^*) . The second forgery therefore satisfies $e(g, \sigma_2^*) = e(U, T'^*) \cdot e(U, X)^{\hat{h}_5^*}$.

Dividing the first verification equation by the second yields $\frac{e(g, \sigma_1^*)}{e(g, \sigma_2^*)} = \frac{e(U, T'^*)e(U, X)^{h_5^*}}{e(U, T'^*)e(U, X)^{\hat{h}_5^*}} = e(U, X)^{h_5^* - \hat{h}_5^*}$. By bilinearity, the left-hand side is $\frac{e(g, \sigma_1^*)}{e(g, \sigma_2^*)} = e(g, \sigma_1^*(\sigma_2^*)^{-1})$. Hence, $e(g, \sigma_1^*(\sigma_2^*)^{-1}) = e(U, X)^{h_5^* - \hat{h}_5^*}$.

Since $X = g^b$, the right-hand side satisfies

$$\begin{aligned} e(U, X)^{h_5^* - \hat{h}_5^*} &= e(U, g^b)^{h_5^* - \hat{h}_5^*} \\ &= e(U^b, g)^{h_5^* - \hat{h}_5^*} = e(g, U^b)^{h_5^* - \hat{h}_5^*} \\ &= e(g, (U^b)^{h_5^* - \hat{h}_5^*}) \end{aligned} \quad (17)$$

where the equality $e(U^b, g) = e(g, U^b)$ follows from the symmetric bilinear setting. Therefore, $e(g, \sigma_1^*(\sigma_2^*)^{-1}) = e(g, (U^b)^{h_5^* - \hat{h}_5^*})$. Therefore, we have $\sigma_1^*(\sigma_2^*)^{-1} = (U^b)^{h_5^* - \hat{h}_5^*}$.

Since $h_5^* \neq \hat{h}_5^*$, the difference $h_5^* - \hat{h}_5^*$ is nonzero in $\mathbb{Z}_q$ and thus has a multiplicative inverse modulo q . Consequently, the challenger computes $U^b = (\sigma_1^*(\sigma_2^*)^{-1})^{(h_5^* - \hat{h}_5^*)^{-1}} = (g^a)^b = g^{ab}$ and therefore solves the given CDH instance with non-negligible probability.

This contradicts the assumed hardness of the CDH problem. Therefore, no PPT adversary can generate a fresh valid authentication response for the target vehicle with non-negligible probability without possessing the corresponding vehicle private key. Consequently, the vehicle authentication response of the proposed heterogeneous identity authentication protocol is existentially unforgeable under chosen-message attacks in the random oracle model.

B. Security Analysis

Theorem 1 establishes the confidentiality of the proposed heterogeneous identity authentication protocol, ensuring that encrypted vehicular traffic information remains protected against unauthorized disclosure. Theorem 2 establishes the unforgeability of authentication messages. In addition to confidentiality and unforgeability, the proposed protocol provides the following security properties.

(1) **DoS Attacks.** The vehicle private-key generation procedure adopts a threshold-based distributed architecture in which multiple RSUs participate in generating partial private keys. The failure or temporary unavailability of a limited number of RSUs therefore does not constitute a single point of failure, provided that a sufficient number of authorized RSUs remain available to complete private-key reconstruction. This design improves the availability and robustness of the authentication service against partial RSU disruption. Network-level volumetric DoS attacks remain outside the protection provided by the cryptographic mechanism itself.

(2) **Replay Attacks.** The protocol incorporates fresh session information together with the random values t_1 and t_i into the authentication procedure. Authentication messages generated in different sessions therefore correspond to different protocol transcripts. Each entity verifies the freshness of the session information and rejects previously used or expired authentication transcripts. Consequently, an adversary cannot successfully authenticate by retransmitting messages captured from an earlier session.

(3) **Internal Attacks.** The threshold private-key generation mechanism prevents an individual RSU from independently deriving a vehicle's complete private key. Compromising fewer RSUs than the required reconstruction threshold is insufficient to recover the master secret or reconstruct the private key of a target vehicle. Moreover, the vehicle authentication response is cryptographically bound to the corresponding vehicle private key. Hence, an authenticated vehicle that obtains session-related information cannot use such information alone to impersonate another legitimate vehicle. Under the assumed trust model for the PKG and CA, the proposed protocol

therefore provides resistance against insider attacks involving compromised vehicles or an insufficient number of RSUs.

(4) **Man-in-the-Middle Attacks.** The incorporation of random values t_1 and t_i prevents adversaries from deriving private keys through eavesdropping. Message integrity is preserved, and unauthorized modification is infeasible, thereby enabling resistance to man-in-the-middle attacks.

(5) **Privacy Protection.** The protocol employs IPFE to protect the privacy of vehicular data during transmission and service-side processing. Vehicles transmit encrypted traffic information rather than plaintext data. The AS obtains only the functional results authorized by the corresponding functional key and does not directly recover the underlying vehicular data. This mechanism preserves the confidentiality of sensitive information while retaining the data-processing capability required by intelligent transportation services.

C. Security Simulation

In addition to the theoretical security analysis, formal verification is conducted using ProVerif to evaluate the specified security properties of the proposed heterogeneous identity authentication protocol under the symbolic adversarial model. The protocol model consists of the following components.

(1) **Declaration Section.** This section defines the variables, constants, communication channels, and cryptographic primitives required by the protocol model. It also specifies the abstract functions and equations used to represent hash operations and the relevant cryptographic computations involved in authentication and key establishment.

(2) **Process Section.** The behaviors of the protocol entities, including vehicles, the CA, the PKG, the AS, and RSUs, are modeled as communicating processes. The simulation instantiates an authentication scenario involving three vehicles and the AS and represents the message exchanges among the participating entities according to the proposed protocol workflow.

(3) **Main Section.** This section defines the concurrent execution of the protocol processes and specifies secrecy queries for security-sensitive values, including the AS private key, vehicle private keys, and the session key shared between the AS and vehicles. These queries determine whether the corresponding secrets can be derived by the adversary under the modeled attack capabilities. The simulation code is publicly available on GitHub¹. As shown in Fig. 2, the first two verification results indicate that the session keys shared between the AS and vehicles remain secret in the formal model. The subsequent results confirm the secrecy of the private keys of the vehicles and the AS. The remaining verification results show that the specified protocol processes satisfy the corresponding ProVerif queries under the adopted symbolic model, providing formal support for the security properties analyzed above.

D. Limitations of Security

CPA Security. The proposed scheme achieves CPA security and therefore does not inherently provide security against

¹<https://github.com/Wj011014/IPFE>

```

Verification summary:
Query not attacker(theta[]) is true.
Query not attacker(theta'[]) is true.
Query not attacker(vsk1[]) is true.
Query not attacker(vsk2[]) is true.
Query not attacker(vsk3[]) is true.
Query not attacker(ask1[]) is true.
Query not attacker(ask2[]) is true.
Query not attacker(ask3[]) is true.
Query not attacker(ask4[]) is true.
Query inj-event(endVehicle(id)) ==> inj-event(beginAS(id)) is true.
Query inj-event(endAS(id)) ==> inj-event(beginVehicle(id)) is true.

```

Fig. 2: The simulation results of the ProVerif

chosen-ciphertext attacks. An adversary with the capability to submit adaptively modified ciphertexts may exploit properties that are not captured by the CPA security model. The authentication mechanisms incorporated into the proposed protocol provide integrity and entity-authentication protection at the protocol level, yet they do not upgrade the underlying CPA-secure encryption construction to CCA security. Consequently, strengthening the ciphertext layer against active chosen-ciphertext attacks remains an important direction for future research.

When stronger protection against chosen-ciphertext attacks is required, a possible extension is to combine a CPA-secure IPFE construction with additional cryptographic mechanisms that provide ciphertext validity and non-malleability, such as non-interactive zero-knowledge proofs and strongly unforgeable one-time signatures. An attribute-based IPFE construction with appropriate CCA-secure schemes may also be incorporated when attribute-set-based functional authorization is required. Such an extension would require corresponding modifications to the ciphertext structure, decryption procedure, and security model, together with a formal proof under the resulting CCA-security definition.

Resistance to Extended Attack Scenarios. For our proposal, beyond conventional cryptographic attacks, several additional threats should be considered. First, our protocol does not provide post-quantum security because its security relies on discrete-logarithm and pairing-based assumptions; sufficiently powerful quantum computers could therefore undermine the underlying cryptographic primitives, and replacing them with post-quantum constructions remains future work. For DoS attacks against RSUs, the threshold key-generation method provides partial resilience because vehicle keys can still be reconstructed as long as at least η authorized RSUs remain available, although large-scale flooding or simultaneous disruption of many RSUs cannot be prevented by the cryptographic protocol alone. Side-channel attacks are not addressed, since we did not implement leakage-resilient protection for the private information. A malicious or compromised AS is restricted by IPFE to obtaining the functional outputs authorized by its functional key rather than the underlying vehicular plaintext; nevertheless, the current protocol does not prevent an authorized server from misusing legitimate computation results or deviating from the prescribed service

logic. Regarding collusion attacks, cooperation among fewer than η compromised RSUs is insufficient to reconstruct a vehicle key, whereas collusion involving sufficiently many RSUs or trusted authorities is outside the current threat model. Finally, certificate revocation and key re-issuance can limit the impact of detected key compromise, while the present construction does not provide formal key-leakage resilience or automatic recovery from long-term private-key exposure. Incorporating proactive key updates and post-compromise recovery mechanisms is therefore an important direction for future work.

VII. ANALYSIS AND COMPARISON OF PERFORMANCE

We present a comprehensive comparative analysis of the proposed privacy-preserving protocol. The performance evaluation considers computational and storage overheads in comparison with inner product functional encryption schemes [23]–[25], and communication overhead in comparison with identity authentication schemes under multi-authenticator environments [26]–[28]. Furthermore, a functional comparison with the above schemes [23]–[28] is provided.

IPFE schemes are selected as benchmarks for computational and storage evaluations because they rely on computationally intensive operations, such as bilinear pairings and discrete logarithms, which impose significant overhead in resource-constrained environments. These schemes aim to achieve efficient encryption and decryption while maintaining strong security guarantees under limited computational resources.

For communication cost evaluation, identity authentication schemes are adopted as reference schemes. These schemes primarily focus on transmission efficiency among communicating entities, and their communication overhead is a key factor in assessing the overall system performance.

A. Comparison of Computation Costs

We analyze the computational efficiency of the proposed scheme by defining the execution times of basic multiplication operations in the multiplicative cyclic groups $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ as M_{g_1} , M_{g_2} , and M_t , respectively. The execution times of exponentiation operations are denoted as E_{g_1} , E_{g_2} , and E_t , respectively. The execution times of bilinear pairing and discrete logarithm operations are denoted as BM and DL , respectively. The vector dimension is denoted by n .

The computational costs of the proposed scheme are compared with those of [23]–[25] across four phases: system initialization (Setup), encryption (Enc), key generation (KeyGen), and decryption (Dec). As shown in Table III, the proposed scheme incurs lower computational overhead in the Setup algorithm than [24], [25]. Although its setup cost is higher than that of [23], the proposed scheme achieves constant computational overhead in encryption, key generation, and decryption, independent of the vector dimension n . In the key generation phase, the proposed scheme also outperforms the fixed-overhead scheme in [25]. Furthermore, its constant ciphertext and key sizes provide additional efficiency advantages.

TABLE III: Comparison results of computation costs between our construction and the existing IPFE scheme

Algorithms	Song <i>et al.</i> [23]	Belel <i>et al.</i> [24]	Han <i>et al.</i> [25]	Our proposed
Setup	$1E_{g_1}$	$(3n+2)E_{g_1} + (n+1)E_t + 1BM$	$2E_{g_1} + (n+1)E_t$	$(n+1)E_{g_1} + BM$
Enc	$(n+2)E_{g_1} + nM_{g_1} + nBM + 2nE_t + nM_t$	$(n+1)E_{g_1} + 2nE_t + 2nM_t + nM_{g_1}$	$(n+6)E_{g_1} + nE_t + nM_{g_1} + nM_t$	$(n+1)E_{g_1} + 1E_t + nM_{g_2}$
KeyGen	$(2n+1)E_{g_1} + nM_{g_1}$	$(2n+1)E_{g_1} + (2n-1)M_{g_1}$	$10E_{g_1} + 4M_{g_1}$	$(n+2)M_{g_1} + 1E_t$
Dec	$2BM + 2M_t + nE_t + 1DL$	$(n+2)E_t + (n-1)M_t + (n+1)BM + 1DL$	$2BM + 2M_t + nE_t + 2DL$	$nM_{g_1} + 2BM$

TABLE IV: Comparison Results of function

Schemes	$ Params $	$ msk $
Song <i>et al.</i> [23]	$(n+d+2) G_1 + 1 G_T $	$1 Z_q^* $
Belel <i>et al.</i> [24]	$(n+d+3) G_1 + 1 G_T $	$n G_1 $
Han <i>et al.</i> [25]	$6 G_1 + 2 G_2 + 1 G_T $	$(n+2) Z_q^* $
Our proposed	$n G_1 + 1 G_T $	$n G_1 $
Schemes	$ CT $	$ sk $
Song <i>et al.</i> [23]	$2 G_1 + n G_T + d Z_q^* $	$(d+2) G_1 + (n+d) Z_q^* $
Belel <i>et al.</i> [24]	$(d+1) G_1 + n G_T + d Z_q^* $	$(d+1) G_1 + (n+d) Z_q^* $
Han <i>et al.</i> [25]	$4 G_1 + 1 G_2 + n G_T $	$2 G_1 + 3 G_2 $
Our proposed	$(n+1) G_1 + 1 G_T $	$2 G_1 + 1 G_T + (n+1) Z_q^* $

To validate the computational performance, experiments were conducted on a PC running Windows 10 with an AMD Ryzen 3-2300U CPU at 2 GHz and 8 GB RAM. The implementation is publicly available on GitHub². The experimental setup considers $d = 10$ vehicles and $t = 8$ partial private keys per vehicle. The vector dimension n ranges from 10 to 50. Four algorithms, namely Setup, Key Generation, Encryption, and Decryption, are evaluated, and the results are illustrated in Fig. 3, Fig. 5, Fig. 4, and Fig. 6.

As shown in Figs. 3–6, the computational costs of the evaluated schemes generally increase with the vector dimension n , reflecting the additional operations required for processing higher-dimensional vectors. However, the proposed scheme exhibits a relatively moderate growth trend across the different phases.

In the setup phase, as illustrated in Fig. 3, the running time of the proposed scheme increases gradually with n and remains comparatively lower than those of the comparison schemes. In particular, [24] exhibits a more pronounced increase in computational cost as the vector dimension grows. A similar trend can be observed in the key generation phase shown in Fig. 4, where the proposed scheme demonstrates a slower growth in running time than [25], indicating favorable scalability with respect to the vector dimension.

For the encryption phase in Fig. 5, the computational cost of all schemes increases as n grows, whereas the proposed scheme maintains a comparatively low running-time trend throughout the evaluated range. In the decryption phase shown in Fig. 6, the running time also generally grows with the vector dimension. Nevertheless, the proposed scheme exhibits a relatively stable and competitive growth trend compared with the other schemes.

Overall, the experimental results show that increasing the vector dimension leads to higher computational costs for most schemes. Compared with the existing schemes, the proposed

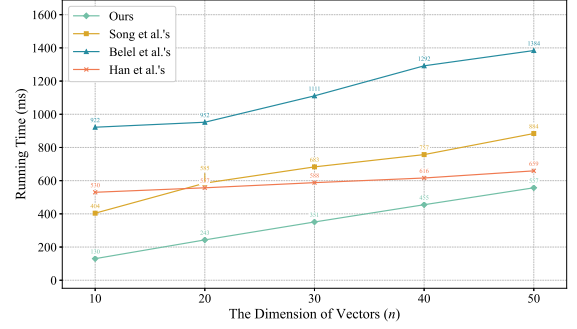


Fig. 3: Computation costs of the Setup algorithm

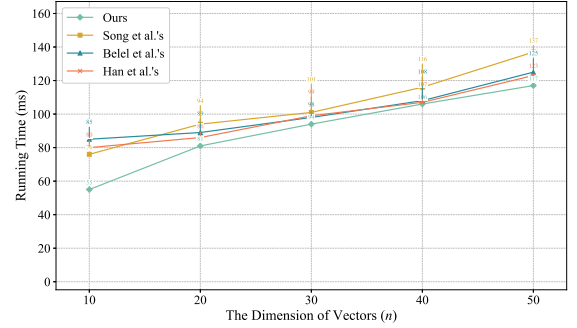


Fig. 4: Computation costs of the Setup algorithm

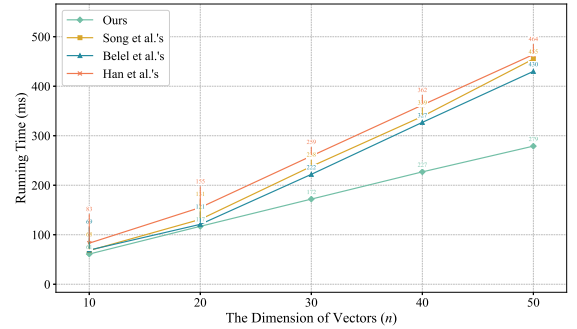


Fig. 5: Computation costs of the Setup algorithm

scheme generally exhibits a more moderate growth trend across the setup, key generation, encryption, and decryption phases. This indicates that the proposed scheme can maintain relatively low computational overhead as the vector dimension increases, demonstrating favorable computational scalability for large-scale VANET applications.

²<https://github.com/Wj011014/IPFE>

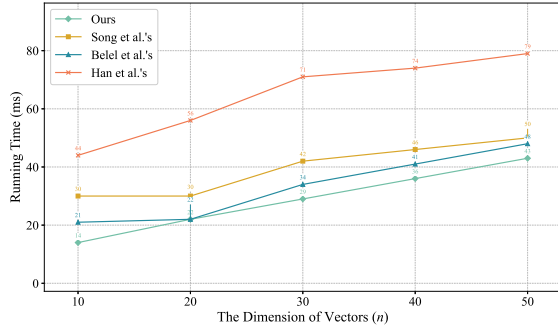


Fig. 6: Computation costs of the Setup algorithm

B. Comparison of Communication Cost

Communication overhead is a critical metric in VANET authentication protocols. We denote the average sizes of elements in $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ as $|G_1|$, $|G_2|$, and $|G_T|$, respectively, and the size of elements in $\mathbb{Z}_q^*$ as $|Z_q^*|$. Under the Type A pairing parameters used in our implementation, the $|Z_q^*|$ is encoded in 20 bytes, while an element of $|G_1|$ occupies 128 bytes using JPBC's default serialization. The communication cost of the proposed scheme is compared with those of [26]–[28], as summarized in Table V.

TABLE V: Comparison Results of Communication cost

Schemes	Single authenticator	Multiple authenticators
scheme [26]	$8 G_1 + 4 Z_q^* $	$8n G_1 + 4n Z_q^* $
scheme [27]	$1 G_1 + 7 Z_q^* $	$n G_1 + 7n Z_q^* $
scheme [28]	$2 G_1 + 3 Z_q^* $	$2n G_1 + 3n Z_q^* $
Our proposed	$4 G_1 + 3 Z_q^* $	$(3n + 1) G_1 + (n + 2) Z_q^* $

As shown in Table V and Fig. 7, the proposed scheme incurs lower communication overhead than [26], but higher overhead than [27], [28] under the adopted parameter settings. Therefore, the proposed scheme does not achieve a clear advantage in communication efficiency, which constitutes a limitation of the current design. Nevertheless, its communication cost grows linearly with the number of authenticators, following the same asymptotic growth order as the comparison schemes.

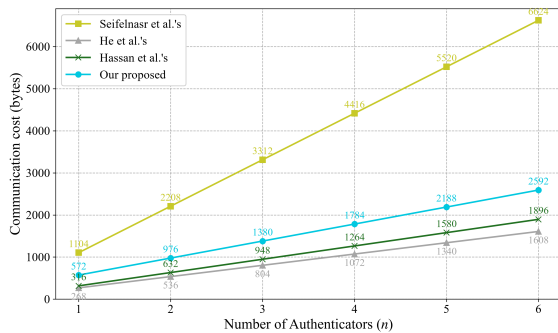


Fig. 7: Comparison Results of Communication Costs

Moreover, the reported communication cost represents the cumulative payload exchanged throughout the entire authentication procedure, rather than the size of a single message. The

total payload is distributed across multiple protocol messages, so each individual transmission carries only a portion of the overall authentication data. It should be noted that this distribution does not reduce the aggregate communication volume and therefore does not eliminate the communication overhead limitation of the proposed scheme. However, it helps distinguish the total communication cost of the authentication procedure from the payload carried by an individual transmission.

Overall, the proposed scheme is not optimal in terms of communication efficiency, and the additional communication overhead represents a practical limitation in deployment. On the other hand, the computational evaluation shows comparatively low running times, while the scheme also supports heterogeneous authentication and privacy-preserving computation over encrypted data. These results therefore reflect a trade-off among communication overhead, computational efficiency, and functionality. In practical applications, the suitability of the proposed scheme depends on whether the additional communication cost can be accommodated in exchange for the corresponding computational and functional capabilities.

C. Comparison of Storage Costs

The storage costs of different schemes are analyzed in Table IV. In [23]–[25], the storage requirement of ciphertext CT increases with both the vector dimension n and the number of users d , resulting in substantial storage overhead in large-scale systems. In particular, the private key size sk in [23], [24] also grows with n and d . Although [25] maintains a fixed private key size, the number of bilinear group elements in the ciphertext grows linearly with n , resulting in higher overall storage cost. Therefore, the advantage of [25] in private key storage is limited in practical deployments. In contrast, the proposed scheme achieves efficient storage overhead for the ciphertext, public parameters, and master secret key. This property significantly reduces storage requirements and enhances scalability for large-scale VANET deployments.

D. Comparison and Analysis of Functions

A functional comparison between the proposed scheme and existing works is presented in Table VI, where “✓” indicates supported features, “✗” indicates unsupported features, and “–” denotes non-applicability.

As shown in Table VI, the schemes in [23]–[25] do not support heterogeneous environments, whereas the proposed scheme does. The schemes in [26], [27] lack fine-grained access control and resistance to internal attacks. Although [6] provides resistance against impersonation and internal attacks, it does not support fine-grained access control. Moreover, none of the schemes in [23]–[29] support multi-receiver authentication.

In contrast, the proposed scheme supports heterogeneous environments, multi-receiver authentication, and fine-grained access control, while also providing resistance against internal and impersonation attacks. These features collectively offer significant advantages in both security and functionality.

TABLE VI: Comparison Results of function

Schemes	Heterogenous	Multi-receiver authentication	Fine-grained access control	Internal attack	Impersonation attack
Song <i>et al.</i> [23]	✗	–	✓	–	–
Belel <i>et al.</i> [24]	✗	–	✓	–	–
Han <i>et al.</i> [25]	✗	–	✓	–	–
Seifelnasr <i>et al.</i> [26]	✗	✗	✗	✗	✓
He <i>et al.</i> [27]	✗	✗	✗	✗	✗
Hassan <i>et al.</i> [28]	✓	✗	✗	✓	✓
Long <i>et al.</i> [29]	✗	✗	✗	✓	✓
Our proposed	✓	✓	✓	✓	✓

E. Comparison of Authentication Method

In conventional VANET authentication schemes, one-to-one authentication requires vehicles to establish multiple independent connections with RSUs, leading to increased communication delay and reduced efficiency. Multiple interactions also enlarge the attack surface.

To address this limitation, the proposed protocol adopts a broadcast authentication mechanism, enabling simultaneous authentication between vehicles and multiple *RSUs*. This parallel authentication process improves efficiency, reduces communication interactions, and lowers overall communication overhead.

A comparison with one-to-one authentication schemes [26]–[28], as shown in Table VII, demonstrates that the proposed scheme significantly improves authentication efficiency, reduces communication cost, and enhances overall system security. Our proposal can support multi-to-one authentication because it uses the inner product as the authentication standard. As a result, the basis for judging identity legitimacy is the result of the inner product calculation, which the system sets to a specific value. If the inner product value of the vector held by the server and the vector held by the user matches the value set by the system, then the user and the server will pass the identity legitimacy authentication. By contrast, traditional one-to-one identity authentication is based on a signature containing the identity, and due to the directionality of the identity, each authentication request can only complete the identity legitimacy authentication with a specific server. Furthermore, in our construction, one-to-many identity authentication is realized through broadcast authentication, allowing a user to authenticate with multiple servers within a single communication round. This design improves authentication efficiency and reduces the computational and communication overhead incurred by users. These properties make the proposed protocol well suited for applications such as the VANETs, where high vehicle mobility requires frequent authentication with multiple RSUs. By enabling simultaneous authentication with multiple RSUs in a single communication round, the proposed protocol facilitates rapid service handover and supports continuous and reliable vehicular communications.

F. Simulation Evaluation in a Realistic Vehicular Environment

The preceding analysis is purely algorithmic: it accounts for the cryptographic primitives but not for the wireless medium, vehicle mobility, or a contended channel. We therefore imple-

ment the proposed protocol in the Veins 5.2 vehicular network simulator and evaluate it under realistic vehicular mobility.

1) Simulation Setup: The simulation couples OMNeT++ 6.0.1 with SUMO 1.18.0 through TraCI, using Veins 5.2. Vehicles and RSUs communicate over IEEE 802.11p (5.89 GHz, 10 MHz, 6 Mbit/s, Veins WAVE stack); each RSU reaches the application server (AS) over a 10 ms one-way wired backhaul, which affects all compared schemes identically. The road network (OpenStreetMap) covers the Yanta campus of Shaanxi Normal University and its surroundings (about 2.0 km×1.7 km), with 25 RSUs placed by a grid-covering heuristic (Fig. 8); Fig. 9 shows the running simulation. Fleets of 20, 50, 100, 200 and 500 vehicles run for 300 s each; the 100-vehicle fleet is also run at 30, 50, 80 and 120 km/h. Demand comes from SUMO's *randomTrips*, with departures over the first 120 s and a minimum trip distance calibrated so that the mean number of vehicles in the network matches the design fleet size; speed limits and signalised junctions are enforced. PHY/MAC parameters are listed in Table VIII.

Cryptographic operations are charged their *measured* cost rather than being treated as instantaneous; the unit costs come from a jPBC 2.0.0 benchmark on the same Type-A parameters as the implementation ($r = 160$ bit, $q = 512$ bit). The attribute vector dimension is $n = 10$, the key-extraction threshold $\eta = 3$, vehicles re-authenticate every 30 s, RSUs aggregate ciphertexts over 5 s, the AS collects requests and responses over 0.5 s and 0.8 s windows, and the brute-force discrete logarithm of the decryption searches 2^{10} elements. All of these steps, from key extraction and combination to RSU aggregation and functional decryption, run as application-layer modules.

Repetitions and Statistics. Every configuration of the proposed scheme is run five times with independent random seeds (fifteen times at the 20-vehicle setting, where run-to-run variation is largest); the competitors are deterministic given the measured primitive costs and run once per fleet size. The seed controls both the SUMO demand generation (92–113 vehicles in the network at the nominal 100-vehicle setting) and the channel/MAC randomness. Reported values are five-run means with 95% confidence intervals (Student's *t* intervals, with the degrees of freedom matched to the number of runs), shown as error bars. When the first 120 s, over which the demand is injected, are excluded from the statistics, the mean latencies change by at most 0.27 s, which is well within the confidence intervals. The simulation model, the SUMO scenario generation and the analysis scripts are provided together with the implementation in the public repository³.

2) Evaluation Results: Now, we will show the evaluation results.

(1) Scalability of the verifier. Fig. 11(a) shows the AS computation per broadcast round against the number of vehicles that respond to the round. In particular, the rounds serving 3.51 ± 0.32 vehicles at 20 vehicles and 38.4 ± 4.6 at 500 cost 30.34 ± 0.01 and 31.13 ± 0.10 ms respectively. This increase is caused by the M_{g_1} operation that the aggregate verification

³<https://github.com/Wj011014/IPFE>

TABLE VII: Comparison Results of authentication methods

Schemes	Authentication Method	Advantage/Disadvantage
Schemes [5], [26], [27], [29]	One-to-One Authentication	1) High resource consumption. 2) Risk of a single point of failure. 3) The authentication process is time-consuming and inefficient. 4) Poor scalability and lack of flexibility.
Our scheme	Broadcast Authentication	1) Low resource consumption. 2) The authentication process is fast and efficient. 3) It has better scalability and flexibility. 4) Reduce communication costs.

TABLE VIII: PHY/MAC parameters of the simulation

Parameter	Value
Channel	5.89 GHz, 10 MHz, 6 Mbit/s (IEEE 802.11p)
Transmit power	100 mW
Receiver sensitivity	-89 dBm
Noise floor	-98 dBm
Path-loss model	log-distance, exponent 2.0
Interference range	1000 m
Obstruction model	none (line-of-sight assumed)
EDCA parameters	CWmin 15, CWmax 1023, AIFS 2/3/6/9

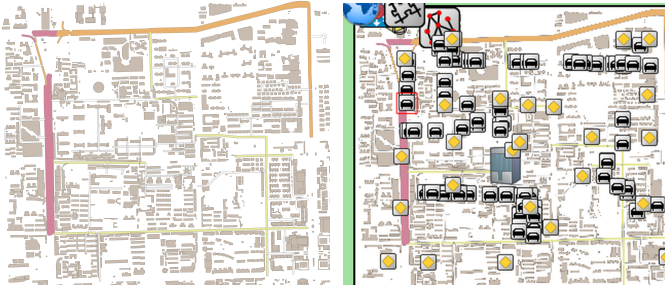
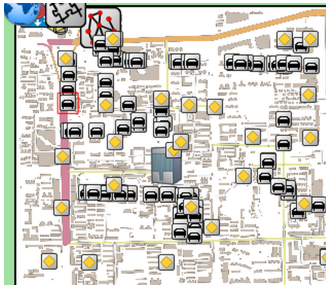


Fig. 8: The road network imported from OpenStreetMap

Fig. 9: Runtime screenshot: 25 RSUs, the AS (centre), and vehicles at $t \approx 119$ s.

performs for each responding vehicle. In addition, about 1–3.3% of the broadcast rounds receive no response within the acknowledgement window and only incur the 13.8 ms delta cost; these rounds are not included in Fig. 11(a), while their cost is contained in the per-vehicle workload of Fig. 11(b). Fig. 11(b) shows the same workload per authenticated vehicle. The one-to-one variant of the proposed scheme consumes 27.9–29.5 ms per vehicle at every fleet size, so its total workload grows linearly with the fleet. The one-to-many mode instead falls from 8.93 ± 0.84 ms at 20 vehicles to 4.33 ± 0.74 , 2.99 ± 0.51 , 1.76 ± 0.33 and 0.83 ± 0.09 ms at 50, 100, 200 and 500 vehicles; at 100 vehicles this is $9.5\times$ less work per vehicle than the one-to-one variant, and $35.6\times$ less at 500. The price of this reduction is a bounded delay. The mean authentication latency increases from 1.58 ± 0.11 s to 2.55 ± 0.04 s, and the end-to-end join time from 3.90 ± 0.75 s to 5.56 ± 0.81 s. Therefore, a fleet 25 times larger is served with about $1.4\times$ the latency.

(2) **Advantage over one-to-many authentication schemes.** Fig. 11(b) also reports three representative one-to-one schemes compared in Section “Comparison of Authentication Method” [26]–[28], each re-implemented as a configuration of the same simulation model, changed the primitive mixture its algorithm

prescribes at the same JPBC prices, with the same message sizes and the same 802.11p stack, mobility traces and RSU deployment. Their per-vehicle workload is fixed at 8.25, 11.0 and 16.5 ms for the light, medium and heavy representatives. The proposed scheme is already cheaper than the medium and heavy representatives at 20 vehicles and undercuts the lightest one from about 50 vehicles on; at 500 vehicles it needs about $10\times$, $13\times$ and $20\times$ less verifier computation per authenticated vehicle, with an authentication message as small as the lightest scheme’s and smaller than the other two (Table V). The competitors perform authentication only, whereas the proposed scheme additionally provides the IPFE functionality and fine-grained access control, so the comparison isolates the authentication mechanism itself.

(3) **Advantage over a batch-verification scheme.** The closest competitor of the broadcast design is a batch-verification scheme, which also verifies a whole cohort in one operation; Fig. 11(b) therefore includes a representative BLS batch scheme in which every vehicle produces one short signature and the AS checks m signatures with $m + 1$ pairings, about one pairing per signature. This is the strongest baseline: at 2.75 ms per authenticated vehicle it beats all three one-to-one schemes, i.e. we charge only the amortised batch cost, while the residual $+1$ pairing per batch and the hash-to- G_1 computation are not included, which favours this baseline. Its per-vehicle cost nevertheless remains nearly constant over the whole fleet range, because the verifier must still process every signature in the pairing product, whereas one broadcast of the proposed scheme authenticates the whole responding cohort at a cost independent of cohort size. The proposed scheme overtakes it once about eleven vehicles share one broadcast (about 110 vehicles in our scenario) and needs $3.3\times$ less verifier computation at 500 vehicles. Below that point the batch baseline is marginally cheaper, but it provides neither the IPFE functionality nor the heterogeneous identity support nor the fine-grained access control of the proposed construction.

(4) **Aggregated Decryption.** Fig. 10 plots the AS decryption delay against the batch size d at 100 vehicles. With component-wise aggregation the delay is constant at 62.16 ± 0.00 ms regardless of d (a deterministic function of the primitive prices), whereas per-ciphertext decryption grows at exactly 62.16 ms per ciphertext and reaches 559 ms at $d = 9$. Over the 1219 windows observed in the five runs the mean batch size is $d = 5.68$, so per-ciphertext decryption would cost 353 ms on average against 62.2 ms for the aggregated mode, a saving factor of 5.7 that grows with the number of vehicles sharing a window. The decryption recovers the inner product by a discrete-logarithm search over 2^{10} values,

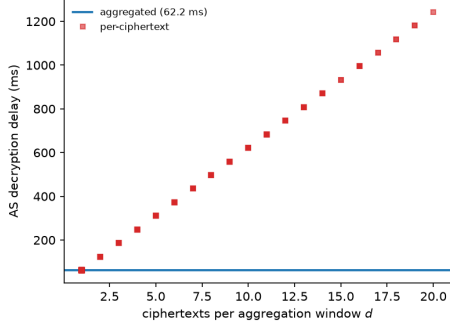


Fig. 10: AS decryption delay vs. batch size d , aggregated vs. per-ciphertext.

so the plaintext is discretised to this range and the cost is linear in the domain size. The domain matches the target analytics, where the inner product stays within 2^{10} ; a larger domain multiplies the search by the same factor (about 60 s for 2^{20}). For unbounded domains, Pollard’s kangaroo reduces the cost to its square root; the aggregation mechanism itself is independent of the domain size.

(5) Deployment Indicators. Virtually every vehicle is authenticated, with a success rate of 100.0% at 20–100 vehicles and $99.9 \pm 0.1\%$ at 200 and 500 vehicles. Collisions do cause losses; a vehicle retransmits up to three times and otherwise retries at the next re-authentication instant, but the mean number of retransmitted attempts falls from 1.61 ± 0.28 per vehicle at 20 vehicles to 0.42 ± 0.10 at 500, i.e. the retransmission overhead of the broadcast decreases as the fleet grows. The authentication message (two $\mathbb{G}_1$ elements plus the identity, 268 B) matches the lightest one-to-one competitor and is smaller than the other two (316 B and 1104 B); only the batch baseline’s signature (140 B) is smaller. The air-interface load reaches 3.07 ± 0.13 Mbit/s at 500 vehicles, within the 6 Mbit/s channel capacity, and is dominated by the periodic ciphertext reports; RSU-side aggregation bounds the backhaul to one aggregated ciphertext per RSU and window. Mobility has little effect: from 30 to 120 km/h the mean join time varies by less than 0.8 s (4.30, 4.05, 4.62 and 3.88 s; Fig. 12(b)). This residual non-monotonicity is caused by the batch composition, and all the differences lie within the 95% confidence intervals. The absolute join time of 3.88–5.56 s is set by the fixed protocol windows: at 100 vehicles, waiting for the aggregation, batch and response windows accounts for 3.15 s of the 4.05 s mean (78%); the growth at large fleets (4.11 s $\rightarrow$ 5.56 s) comes from air-interface contention and AS queueing, which take over as windowing drops to 57% of the total at 500 vehicles. Such latency suits the targeted periodic-reporting and encrypted-analytics services; sub-second safety-critical budgets would require shorter aggregation windows, trading aggregation efficiency for latency (quantified below).

(6) Parameter Sensitivity. Sweeps on the 100-vehicle scenario (single runs, indicative) show that the aggregation window W only changes how many ciphertexts share a decryption: for $W = 1, 5, 10, 20$ s the AS performs 599, 219, 126 and 72 decryptions of 2.1–16.2 ciphertexts, each at 62.2 ms,

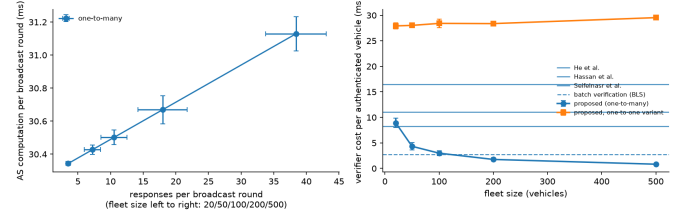


Fig. 11: AS verifier cost (mean per configuration, error bars: 95% confidence interval; 15 runs at 20 vehicles, 5 elsewhere): (a) per broadcast round with at least one response; (b) per authenticated vehicle vs. fleet size. The one-to-one and batch-verification competitor curves authenticate only and do not provide the IPFE functionality.

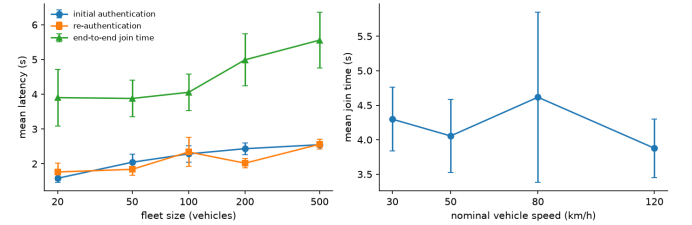


Fig. 12: Mean latency vs. fleet size (a) and vs. vehicle speed (b); error bars show the 95% confidence interval (15 runs at 20 vehicles, 5 elsewhere).

with identical authentication outcomes. $W = 5\text{--}20$ s keeps decryption work at or below the authentication work (13.6–4.5 s against 16.7 s), whereas $W = 1$ s would make it dominant (37 s). The re-authentication interval R scales the load roughly as $1/R$: $R = 10, 30, 60$ s yield 1401, 594 ± 177 and 312 successful authentications at 100 vehicles. Therefore, we recommend W of 5–20 s, and R can be selected according to the freshness requirement of the service.

(7) Infrastructure Density. Re-running the 100- and 500-vehicle scenarios with 15 and 35 RSUs leaves the mechanism unchanged: one broadcast still costs 29.8–30.8 ms across all densities, because it depends on the batch size rather than on the number of forwarding RSUs. Denser infrastructure slightly reduces the fan-out (10.8 to 9.4 vehicles per broadcast at 100 vehicles; 39.4 to 35.6 at 500) and raises the per-vehicle cost correspondingly (2.76–3.21 ms at 100 vehicles), while latency (3.64–4.38 s at 100 vehicles, 4.55–5.56 s at 500) and the success rate (100%, 99.9–100%) are insensitive. Therefore, the advantage of the one-to-many authentication over one-to-one verification is inherent to the scheme and does not depend on the particular deployment.

3) Discussion: In summary, the simulation with five seeds per configuration and 95% confidence intervals supports the following conclusions. (1) The measured primitive costs translate into bounded end-to-end latencies, which are set by the protocol windows rather than by cryptography or the wireless medium. (2) Broadcast one-to-many authentication lowers the AS computation per authenticated vehicle as the fleet grows, whereas one-to-one and batch verification cost the same per vehicle at every size. Therefore, beyond about 110 vehicles

the proposed design requires the least verification computation among all the compared schemes. (3) Ciphertext aggregation keeps both the AS decryption cost and the backhaul traffic independent of the batch size. (4) The protocol works stably for vehicle speeds from 30 to 120 km/h and RSU densities from 15 to 35, and still operates normally at the channel utilisation reached by the largest fleets. Overall, the proposed protocol requires less verification computation than one-to-one and batch authentication in large fleets, and it is the only compared scheme that additionally provides the IPFE functionality and fine-grained access control. These results indicate that the proposed protocol is practical for large-scale urban vehicular environments.

G. Practical Deployment Considerations in VANETs

The practical deployment of the proposed hybrid PKI-IBC architecture requires coordinated key and trust management among vehicles, RSUs, the PKG, the CA, and application servers. On the vehicle side, identity-based private keys can be generated with the assistance of authorized RSUs, reducing the certificate-management burden for highly mobile vehicles. On the AS side, PKI certificates provide explicit trust binding between server identities and public keys. In practice, compromised or expired AS certificates should be revoked through existing PKI revocation mechanisms, such as certificate revocation lists or short-lived certificates, while vehicle key revocation can be supported by periodically updating identity-related key material or incorporating validity periods into key issuance. Interoperability across administrative domains additionally requires mutual recognition of trust authorities or secure cross-domain validation of CA and PKG credentials. Practical deployment must also consider RSU availability, authority synchronization, revocation-information dissemination, and intermittent vehicular connectivity, which may introduce additional communication and management overhead in large-scale intelligent transportation systems.

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